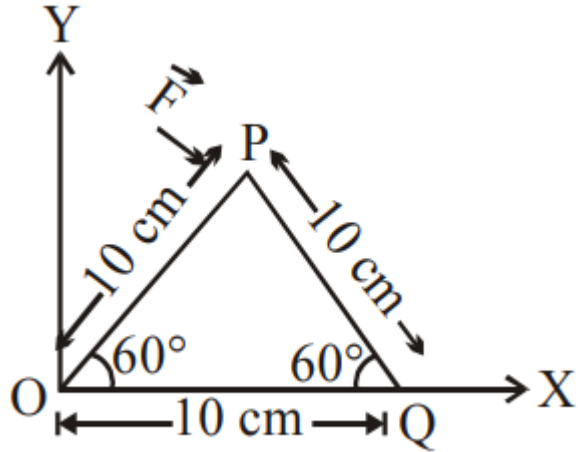
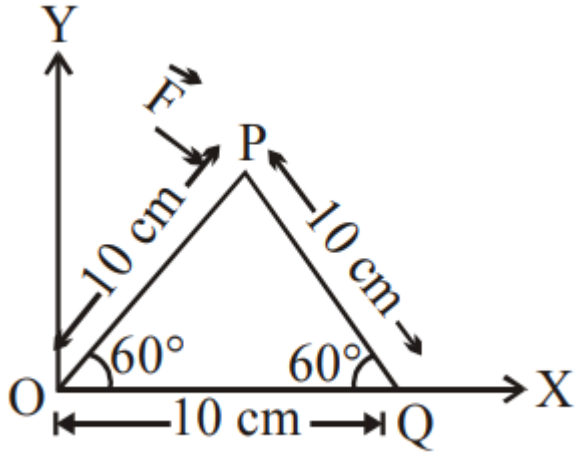


Q1: A triangle plate is shown $\vec{F} = 4\hat{i} - 3\hat{j}$ is applied at point P. The torque at point P with respect to point 'O' and 'Q' are:



- (A) $-15 - 20\sqrt{3}, 15 - 20\sqrt{3}$
- (B) $+15 + 20\sqrt{3}, 15 - 20\sqrt{3}$
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Solution:

$$\vec{F} = 4\hat{i} - 3\hat{j}$$

$$\vec{r}_1 = 5\hat{i} + 5\sqrt{3}\hat{j} \text{ \& } \vec{r}_2 = -5\hat{i} + 5\sqrt{3}\hat{j}$$

Torque about 'O'

$$\vec{\tau}_O = \vec{r}_1 \times \vec{F} = (-15 - 20\sqrt{3})\hat{k} = (15 + 20\sqrt{3})(-\hat{k})$$

Torque about 'Q'

$$\vec{\tau}_Q = \vec{r}_2 \times \vec{F} = (15 - 20\sqrt{3})\hat{k}$$

Q2: When two soap bubbles of radii a and b ($b > a$) coalesce, the radius of curvature of common surface is:

(A) $\frac{ab}{b-a}$

(B) $\frac{a+b}{ab}$

(C) $\frac{b-a}{ab}$

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Solution:

Excess pressure at common surface is given by

$$P_{ex} = 4T \left(\frac{1}{a} - \frac{1}{b} \right) = \frac{4T}{r}$$

$$\therefore \frac{1}{r} = \frac{1}{a} - \frac{1}{b}$$

$$r = \frac{ab}{b-a}$$

Q3: A polyatomic ideal gas has 24 vibrational modes. What is the value of γ ?

- (A) 1.03
- (B) 1.30
- (C) 1.37
- (D) 10.3

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(A) 1.03

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(D) 10.3

Solution:

Since each vibrational modes has 2 degrees of freedom hence total vibrational degrees of freedom = 48

$$f = 3(T) + 3(R) + 24 \times 2(V) = 54$$

$$f = 3 + 3 + 48 = 54$$

$$\gamma = 1 + \frac{2}{f} = \frac{28}{27} = 1.03$$

Q4: If an electron is moving in the n^{th} orbit of the hydrogen atom, then its velocity (v_n) for the n^{th} orbit is given as:

(A) $v_n \propto n$

(B) $v_n \propto \frac{1}{n}$

(C) $v_n \propto n^2$

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(D) $v_n \propto \frac{1}{n^2}$

Solution:

We know velocity of electron in n^{th} shell of hydrogen atom is given by

$$v = \frac{2\pi kZe^2}{nh}$$

$$\therefore v \propto \frac{1}{n}$$

Q5: An electron of mass m and a photon have same energy E . The ratio of wavelength of electron to that photon is: (c being the velocity of light)

(A) $\frac{1}{c} \left(\frac{2m}{E} \right)^{1/2}$

(B) $\frac{1}{c} \left(\frac{E}{2m} \right)^{1/2}$

(C) $\left(\frac{E}{2m} \right)^{1/2}$

(D) $c(2mE)^{1/2}$

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(D) $c(2mE)^{1/2}$

Solution:

$$\lambda_1 = \frac{h}{\sqrt{2mE}}$$

$$\lambda_2 = \frac{hc}{E}$$

$$\frac{\lambda_1}{\lambda_2} = \frac{1}{c} \left(\frac{E}{2m} \right)^{1/2}$$

Q6: Two identical metal wires of thermal conductivities K_1 and K_2 respectively are connected in series. The effective thermal conductivity of the combination is:

(A) $\frac{2K_1K_2}{K_1+K_2}$

(B) $\frac{K_1+K_2}{2K_1K_2}$

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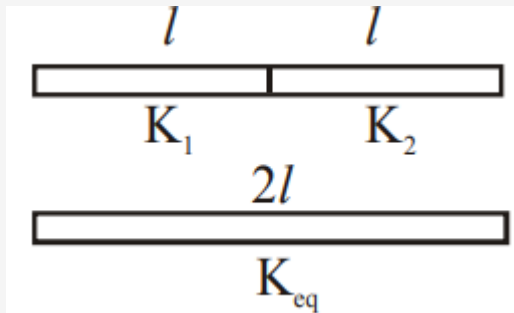
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(C) $\frac{K_1+K_2}{K_1K_2}$

(D) $\frac{K_1K_2}{K_1+K_2}$

Solution:



$$R_{eff} = \frac{l}{K_1 A} + \frac{l}{K_2 A} = \frac{2l}{K_{eq} A}$$

$$K_{eq} = \frac{2K_1K_2}{K_1 + K_2}$$

Q7: The Vernier scale used for measurement has a positive zero error of 0.2 mm. If while taking a measurement it was noted that '0' on the Vernier scale lies between 8.5 cm and 8.6 cm, Vernier coincidence is 6, then the correct value of measurement is _____ cm.
(least count = 0.01 cm)

- (A) 8.36 cm
- (B) 8.54 cm
- (C) 8.58 cm
- (D) 8.56 cm

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(least count = 0.01 cm)

- (A) 8.36 cm
- ☒ (B) 8.54 cm
- (C) 8.58 cm
- (D) 8.56 cm

Solution:

Positive zero error = 0.2 mm

Main scale reading = 8.5 cm

Vernier scale reading = $6 \times 0.01 = 0.06 \text{ cm}$

Final reading = $8.5 + 0.06 - 0.02 = 8.54 \text{ cm}$.

Q8: An AC current is given by $I = I_1 \sin \omega t + I_2 \cos \omega t$. A hot wire ammeter will give a reading:

(A) $\sqrt{\frac{I_1^2 - I_2^2}{2}}$

(B) $\sqrt{\frac{I_1^2 + I_2^2}{2}}$

(C) $\frac{I_1 + I_2}{\sqrt{2}}$

(D) $\frac{I_1 + I_2}{2\sqrt{2}}$

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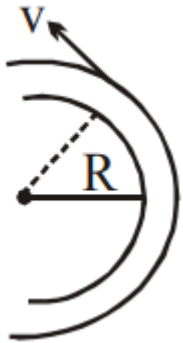
Solution:

$$I = I_1 \sin \omega t + I_2 \cos \omega t$$

$$\therefore I_0 = \sqrt{I_1^2 + I_2^2}$$

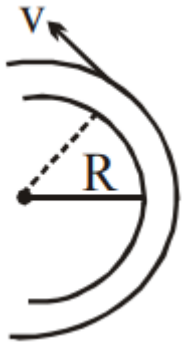
$$\therefore I_{rms} = \frac{I_0}{\sqrt{2}} = \sqrt{\frac{I_1^2 + I_2^2}{2}}$$

Q9: A modern grand-prix racing car of mass m is travelling on a flat track in a circular arc of radius R with a speed v . If the coefficient of static friction between the tyres and the track is μ_s , then the magnitude of negative lift F_L acting downward on the car is:
 (Assume forces on the four tyres are identical and g = acceleration due to gravity)



- (A) $m \left(\frac{v^2}{\mu_s R} + g \right)$
- (B) $m \left(\frac{v^2}{\mu_s R} - g \right)$
- (C) $m \left(g - \frac{v^2}{\mu_s R} \right)$
- (D) $-m \left(g + \frac{v^2}{\mu_s R} \right)$

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- (D) $-m \left(g + \frac{v^2}{\mu_s R} \right)$

Solution:

$$\mu_S N = \frac{mv^2}{R}$$

$$N = \frac{mv^2}{\mu_S R} = mg + F_L$$

$$F_L = \frac{mv^2}{\mu_S R} - mg$$

Q10: A car accelerates from rest at a constant rate α for some time after which it decelerates at a constant rate β to come to rest. If the total time elapsed is t seconds, the total distance travelled is:

(A) $\frac{4\alpha\beta}{(\alpha+\beta)}t^2$

(B) $\frac{2\alpha\beta}{(\alpha+\beta)}t^2$

(C) $\frac{\alpha\beta}{2(\alpha+\beta)}t^2$

(D) $\frac{\alpha\beta}{4(\alpha+\beta)}t^2$

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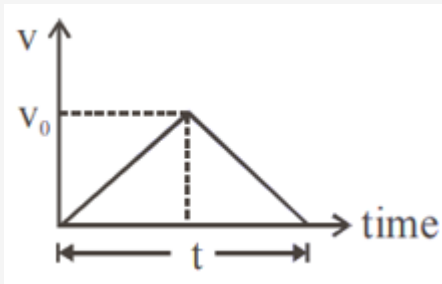
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(C) $\frac{\alpha\beta}{2(\alpha+\beta)}t^2$

(D) $\frac{\alpha\beta}{4(\alpha+\beta)}t^2$

Solution:



$$v_0 = \alpha t_1 \text{ and } 0 = v_0 - \beta t_2 \Rightarrow v_0 = \beta t_2$$
$$t_1 + t_2 = t$$

$$v_0 \left(\frac{1}{\alpha} + \frac{1}{\beta} \right) = t$$

$$\Rightarrow v_0 = \frac{\alpha \beta t}{\alpha + \beta}$$

Distance = area of $v - t$ graph

$$= \frac{1}{2} \times t \times v_0 = \frac{1}{2} \times t \times \frac{\alpha \beta t}{\alpha + \beta} = \frac{\alpha \beta t^2}{2(\alpha + \beta)}$$

Q11: A solenoid of 1000 turns per metre has a core with relative permeability 500. Insulated windings of the solenoid carry an electric current of 5 A. The magnetic flux density produced by the solenoid is:

(Permeability of free space = $4\pi \times 10^{-7} \text{ H/m}$)

- (A) $\pi \text{ T}$
- (B) $2 \times 10^{-3} \pi \text{ T}$
- (C) $\frac{\pi}{5} \text{ T}$
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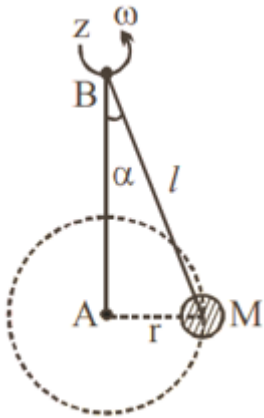
Solution:

$$B = \mu n I = \mu_0 \mu_r n I$$

$$B = 4\pi \times 10^{-7} \times 500 \times 1000 \times 5$$

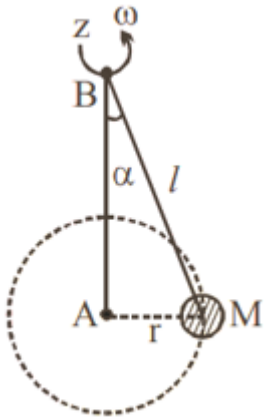
$$B = \pi \text{ Tesla}$$

Q12: A mass M hangs on a massless rod of length l which rotates at a constant angular frequency. The mass M moves with steady speed in a circular path of constant radius. Assume that the system is in steady circular motion with constant angular velocity ω . The angular momentum of M about point A is $\mathbf{L}_A$ which lies in the positive z direction and the angular momentum of M about B is $\mathbf{L}_B$. The correct statement for this system is:



- (A) $\mathbf{L}_A$ and $\mathbf{L}_B$ are both constant in magnitude and direction
- (B) $\mathbf{L}_B$ is constant in direction with varying magnitude
- (C) $\mathbf{L}_B$ is constant, both in magnitude and direction
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Solution:

We know, $\vec{L} = m (\vec{r} \times \vec{v})$

Now with respect to A, we always get direction of $\vec{L}$ along +ve z-axis and also constant magnitude as mvr . But with respect to B, we get constant magnitude but continuously changing direction.

Q13: For what value of displacement the kinetic energy and potential energy of a simple harmonic oscillation become equal?

(A) $x = 0$

(B) $x = \pm A$

(C) $x = \pm \frac{A}{\sqrt{2}}$

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Solution:

K.E = P.E

$$\frac{1}{2}m\omega^2 (A^2 - x^2) = \frac{1}{2}m\omega^2 x^2$$

$$A^2 - x^2 = x^2$$

$$2x^2 = A^2$$

$$x = \pm \frac{A}{\sqrt{2}}$$

Q14: A Carnot's engine working between 400 K and 800 K has a work output of 1200 J per cycle. The amount of heat energy supplied to the engine from the source in each cycle is:

- (A) 3200 J
- (B) 1800 J
- (C) 1600 J
- (D) 2400 J

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Solution:

$$\frac{T_2}{T_1} = \frac{Q_2}{Q_1} = \frac{Q_1 - W}{Q_1} \quad (\because W = Q_1 - Q_2)$$

$$\frac{400}{800} = 1 - \frac{W}{Q_1}$$

$$\frac{W}{Q_1} = 1 - \frac{1}{2} = \frac{1}{2}$$

$$Q_1 = 2W = 2400 \text{ J}$$

Q15: The thickness at the centre of a plano convex lens is 3 mm and the diameter is 6 cm. If the speed of light in the material of the lens is $2 \times 10^8 \text{ m/s}$. The focal length of the lens is

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- (B) 15 cm
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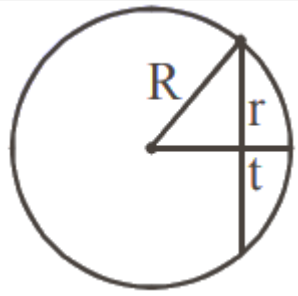
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Solution:

$$R^2 = r^2 + (R - t)^2$$

$$R^2 = r^2 + R^2 + t^2 - 2Rt$$

Neglecting t^2 , we get



$$R = \frac{r^2}{2t}$$

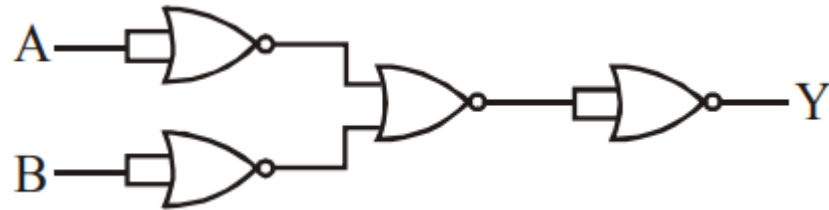
$$\therefore \frac{1}{f} = (\mu - 1) \left(\frac{1}{R} - \frac{1}{\infty} \right) = \frac{\mu - 1}{R}$$

$$f = \frac{R}{\mu - 1} = \frac{r^2}{2t(\mu - 1)} = \frac{(3 \times 10^{-2})^2}{2 \times 3 \times 10^{-3} \times \left(\frac{3}{2} - 1 \right)}$$

$$= \frac{9 \times 10^{-4}}{6 \times 10^{-3} \times 1} \times 2$$

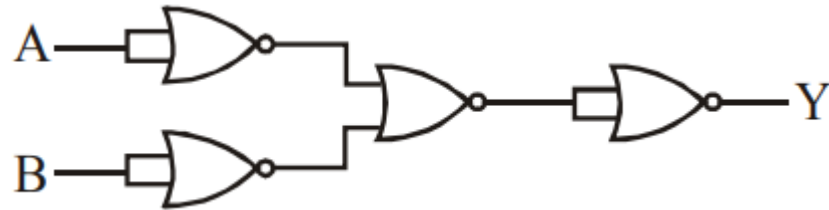
$$f = 0.3 \text{ m} = 30 \text{ cm}$$

Q16: The output of the given combination gates represents:



- (A) XOR Gate
- (B) NAND Gate
- (C) AND Gate
- (D) NOR Gate

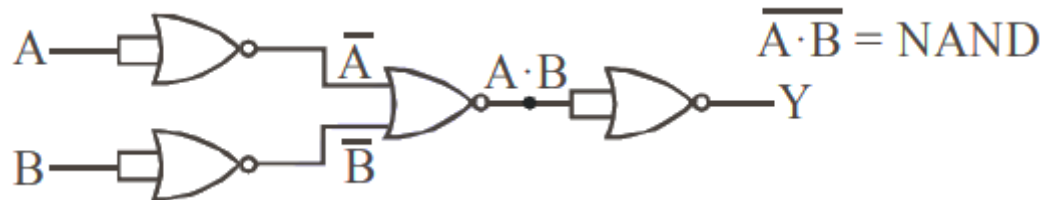
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Solution:

By De Morgan's theorem, we have



Q17: A boy is rolling a 0.5 kg ball on the frictionless floor with the speed of 20 m/s. The ball gets deflected by an obstacle on the way. After deflection it moves with 5% of its initial kinetic energy. What is the speed of the ball now?

- (A) 19.0 m/s
- (B) 4.47 m/s
- (C) 14.41 m/s
- (D) 1.00 m/s

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Solution:

Given, $m = 0.5 \text{ kg}$ and $u = 20 \text{ m/s}$

Initial kinetic energy (k_i) = $\frac{1}{2}mu^2$

$$= \frac{1}{2} \times 0.5 \times 20 \times 20 = 100 \text{ J}$$

After deflection it moves with 5% of k_i

$$\begin{aligned}\therefore k_f &= \frac{5}{100} \times k_i \Rightarrow \frac{5}{100} \times 100 \\ &\Rightarrow k_f = 5 \text{ J}\end{aligned}$$

Now, let the final speed be $v \text{ m/s}$, then

$$k_f = 5 = \frac{1}{2}mv^2$$

$$\Rightarrow v^2 = 20$$

$$\Rightarrow v = \sqrt{20} = 4.47 \text{ m/s}$$

Q18: Which level of the single ionized carbon has the same energy as the ground state energy of hydrogen atom?

- (A) 1
- (B) 6
- (C) 4
- (D) 8

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(A) 1

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(C) 4

(D) 8

Solution:

Energy of H-atom is $E = -13.6 Z^2/n^2$

For H-atom $Z = 1$ & for ground state, $n = 1$

$$\Rightarrow E = -13.6 \times \frac{1^2}{1^2} = -13.6 \text{ eV}$$

Now for carbon atom (single ionized), $Z = 6$

$$E = -13.6 \frac{Z^2}{n^2} = -13.6 \text{ (given)}$$

$$\Rightarrow n^2 = 6^2 \Rightarrow n = 6$$

Q19: Two ideal polyatomic gases at temperatures T_1 and T_2 are mixed so that there is no loss of energy. If F_1 and F_2 , m_1 and m_2 , n_1 and n_2 be the degrees of freedom, masses, number of molecules of the first and second gas respectively, the temperature of mixture of these two gases is:

- (A) $\frac{n_1 T_1 + n_2 T_2}{n_1 + n_2}$
- (B) $\frac{n_1 F_1 T_1 + n_2 F_2 T_2}{n_1 F_1 + n_2 F_2}$
- (C) $\frac{n_1 F_1 T_1 + n_2 F_2 T_2}{F_1 + F_2}$
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- (C) $\frac{n_1 F_1 T_1 + n_2 F_2 T_2}{F_1 + F_2}$
- (D) $\frac{n_1 F_1 T_1 + n_2 F_2 T_2}{n_1 + n_2}$

Solution:

Let the final temperature of the mixture be T . Since, there is no loss in energy.

$$\Delta U = 0$$

$$\Rightarrow \frac{F_1}{2} n_1 R \Delta T + \frac{F_2}{2} n_2 R \Delta T = 0$$

$$\Rightarrow \frac{F_1}{2} n_1 R (T_1 - T) + \frac{F_2}{2} n_2 R (T_2 - T) = 0$$

$$\Rightarrow T = \frac{F_1 n_1 R T_1 + F_2 n_2 R T_2}{F_1 n_1 R + F_2 n_2 R} \Rightarrow \frac{F_1 n_1 T_1 + F_2 n_2 T_2}{F_1 n_1 + F_2 n_2}$$

Q20: A current of 10 A exists in a wire of cross-sectional area of 5 mm^2 with a drift velocity of $2 \times 10^{-3} \text{ m/s}$. The number of free electrons in each cubic meter of the wire is _____.

- (A) 2×10^6
- (B) 625×10^{25}
- (C) 2×10^{25}
- (D) 1×10^{23}

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- (D) 1×10^{23}

Solution:

$$i = 10 \text{ A}, A = 5 \text{ mm}^2 = 5 \times 10^{-6} \text{ m}^2$$

$$\text{and } v_d = 2 \times 10^{-3} \text{ m/s}$$

We know, $i = neAv_d$

$$\begin{aligned} \therefore 10 &= n \times 1.6 \times 10^{-19} \times 5 \times 10^{-6} \times 2 \times 10^{-3} \\ \Rightarrow n &= 0.625 \times 10^{28} = 625 \times 10^{25} \end{aligned}$$

Q21: For VHF signal broadcasting, _____ km^2 of maximum service area will be covered by an antenna tower of height 30 m, if the receiving antenna is placed at ground. Let radius of the earth be 6400 km. (Round off to the Nearest Integer) (Take π as 3.14)

Q21: For VHF signal broadcasting, _____ km^2 of maximum service area will be covered by an antenna tower of height 30 m, if the receiving antenna is placed at ground. Let radius of the earth be 6400 km. (Round off to the Nearest Integer) (Take π as 3.14)

1206.00

Solution:

$$d = \sqrt{2RH}$$

$$A = \pi d^2$$

$$A = \pi 2Rh$$

$$= 3.14 \times 2 \times 6400 \times \frac{30}{1000}$$

$$A = 1205.76 \text{ km}^2$$

$$A \simeq 1206 \text{ km}^2$$

Q22: The angular speed of truck wheel is increased from 900 rpm to 2460 rpm in 26 seconds. The number of revolutions by the truck engine during this time is_____.

(Assuming the acceleration to be uniform).

Q22: The angular speed of truck wheel is increased from 900 rpm to 2460 rpm in 26 seconds. The number of revolutions by the truck engine during this time is _____.
(Assuming the acceleration to be uniform).

Solution:

We know, $\theta = \left(\frac{\omega_1 + \omega_2}{2} \right) t$

Let number of revolutions be N

$$\therefore 2\pi N = 2\pi \left(\frac{900 + 2460}{60 \times 2} \right) \times 26$$
$$N = 728$$

Q23: The equivalent resistance of series combination of two resistors is ' s '. When they are connected in parallel, the equivalent resistance is ' p '. If $s = np$, then the minimum value for n is _____. (Round off to the Nearest Integer)

Q23: The equivalent resistance of series combination of two resistors is ' s '. When they are connected in parallel, the equivalent resistance is ' p '. If $s = np$, then the minimum value for n is _____. (Round off to the Nearest Integer)

4.00

Solution:

$$R_1 + R_2 = s \dots (1)$$

$$\frac{R_1 R_2}{R_1 + R_2} = p \dots (2)$$

As in question $s = np$

$$R_1 + R_2 = \frac{n R_1 R_2}{(R_1 + R_2)}$$

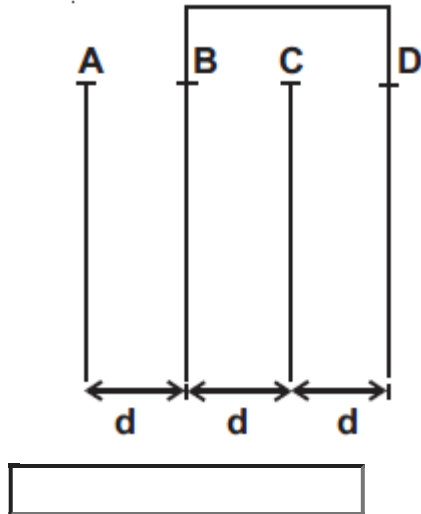
$$\frac{(R_1 + R_2)^2}{R_1 R_2} = n$$

For minimum value of n

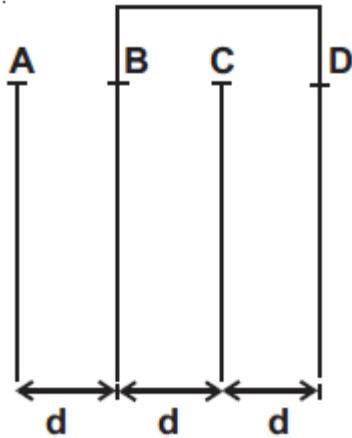
$$R_1 = R_2 = R$$

$$\therefore n = \frac{(2R)^2}{R^2} = 4$$

Q24: Four identical rectangular plates with length, $l = 2\text{ m}$ and breadth, $b = \frac{3}{2}\text{ m}$ are arranged as shown in figure. The equivalent capacitance between A and C is $\frac{x\epsilon_0}{d}$. The value of x is _____. (Round off to the Nearest Integer)

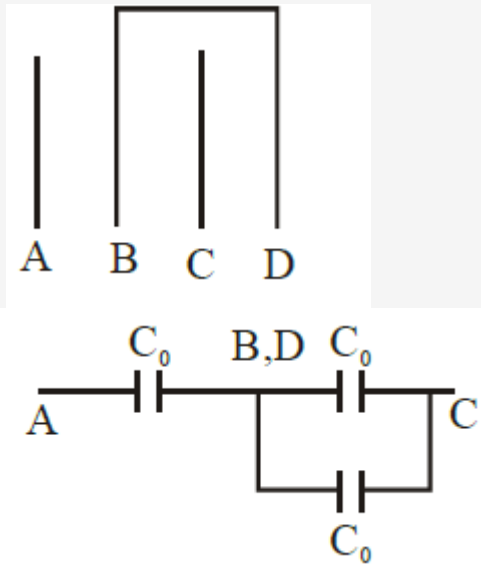


Q24: Four identical rectangular plates with length, $l = 2\text{ m}$ and breadth, $b = \frac{3}{2}\text{ m}$ are arranged as shown in figure. The equivalent capacitance between A and C is $\frac{x\epsilon_0}{d}$. The value of x is _____. (Round off to the Nearest Integer)



2.00

Solution:



$$C_{eq} = \frac{2C_0}{3} = \frac{2}{3} \frac{\epsilon_0 A}{d}$$

$$C_{eq} = \frac{2\epsilon_0}{3d} \times \left(2 \times \frac{3}{2}\right) = 2\epsilon_0 \left(\because A = l \cdot b = 2 \times \frac{3}{2}\right)$$

Q25: The radius in kilometer to which the present radius of earth ($R = 6400$ km) to be compressed, so that the escape velocity is increased 10 times is_____.

Q25: The radius in kilometer to which the present radius of earth ($R = 6400 \text{ km}$) to be compressed, so that the escape velocity is increased 10 times is_____.

64.00

Solution:

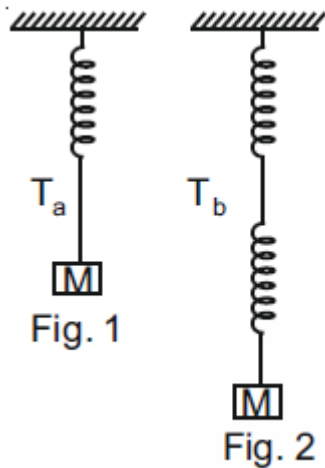
$$V_e = \sqrt{\frac{2Gm}{R}} \dots (1)$$

$$10 V_e = \sqrt{\frac{2Gm}{R'}} \dots (2)$$

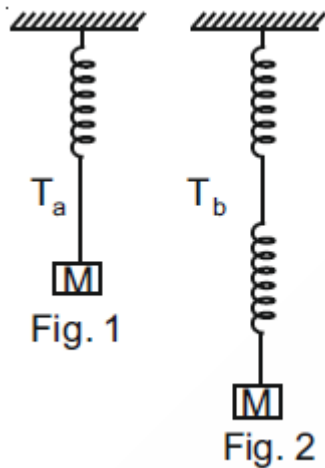
$$\therefore 10 = \sqrt{\frac{R}{R'}}$$

$$\Rightarrow R' = \frac{R}{100} = \frac{6400}{100} = 64 \text{ km}$$

Q26: Consider two identical springs each of spring constant k and negligible mass compared to the mass M as shown. Figure 1 shows one of them and figure 2 shows their series combination. The ratios of time period of oscillation of the two SHM is $\frac{T_b}{T_a} = \sqrt{x}$, where value of x is _____. (Round off to the Nearest Integer)



Q26: Consider two identical springs each of spring constant k and negligible mass compared to the mass M as shown. Figure 1 shows one of them and figure 2 shows their series combination. The ratios of time period of oscillation of the two SHM is $\frac{T_b}{T_a} = \sqrt{x}$, where value of x is _____. (Round off to the Nearest Integer)



2.00

Solution:

$$T_a = 2\pi\sqrt{\frac{M}{K}}$$

$$T_b = 2\pi\sqrt{\frac{M}{K/2}}$$

$$\frac{T_b}{T_a} = \sqrt{2} = \sqrt{x}$$

$$\Rightarrow x = 2$$

Q27: The following bodies,

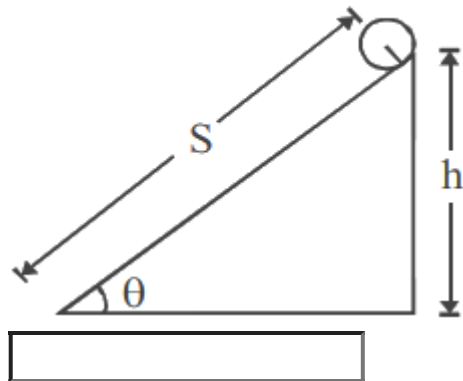
(1) a ring

(2) a disc

(3) a solid cylinder

(4) a solid sphere,

of same mass ' m ' and radius ' R ' are allowed to roll down without slipping simultaneously from the top of the inclined plane. The body which will reach first at the bottom of the inclined plane is _____. [Mark the body as per their respective numbering given in the question]



Q27: The following bodies,

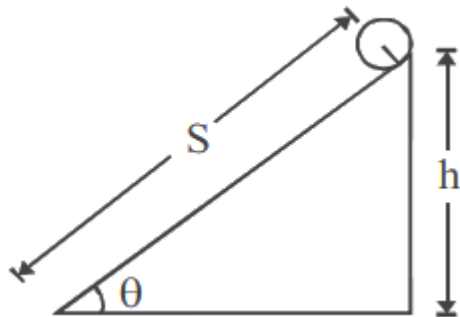
(1) a ring

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4.00

Solution:

$$mg \sin \theta R = (mk^2 + mR^2) \alpha$$

$$\alpha = \frac{Rg \sin \theta}{k^2 + R^2} \Rightarrow a = \frac{g \sin \theta}{1 + \frac{k^2}{R^2}}$$

$$t = \sqrt{\frac{2s}{a}} = \sqrt{\frac{2s}{g \sin \theta} \left(1 + \frac{k^2}{R^2}\right)}$$

For least time, k should be least & we know k is least for solid sphere.

Q28: A parallel plate capacitor whose capacitance C is 14 pF is charged by a battery to a potential difference $V = 12$ V between its plates. The charging battery is now disconnected and a porcelain plate with $k = 7$ is inserted between the plates, then the plate would oscillate back and forth between the plates with a constant mechanical energy of _____ pJ.

(Assume no friction)

Q28: A parallel plate capacitor whose capacitance C is 14 pF is charged by a battery to a potential difference $V = 12$ V between its plates. The charging battery is now disconnected and a porcelain plate with $k = 7$ is inserted between the plates, then the plate would oscillate back and forth between the plates with a constant mechanical energy of _____ pJ.

(Assume no friction)

864.00

Solution:

$$U_i = \frac{1}{2} \times 14 \times 12 \times 12 \text{ pJ } (\because U = \frac{1}{2} CV^2) \\ = 1008 \text{ pJ}$$

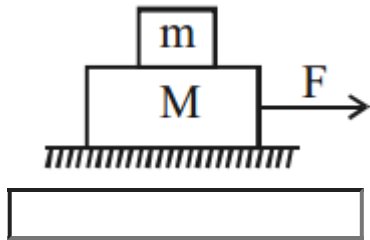
$$U_f = \frac{1008}{7} \text{ pJ} = 144 \text{ pJ } (\because C_m = kC_0)$$

$$\text{Mechanical energy} = \Delta U$$

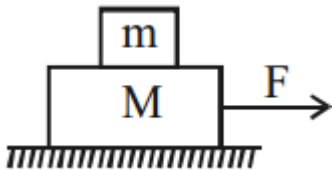
$$= 1008 - 144$$

$$= 864 \text{ pJ}$$

Q29: Two blocks ($m = 0.5 \text{ kg}$ and $M = 4.5 \text{ kg}$) are arranged on a horizontal frictionless table as shown in figure. The coefficient of static friction between the two blocks is $3/7$. Then the maximum horizontal force that can be applied on the larger block so that the blocks move together is _____ N. (Round off to the Nearest Integer) (Take g as 9.8 m/s^2)



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21.00

Solution:

$$a_{\max} = \mu g = \frac{3}{7} \times 9.8$$

$$F = (M + m) a_{\max} = 5a_{\max} \\ = 21 \text{ Netwon}$$

Q30: If $2.5 \times 10^{-6} \text{ N}$ average force is exerted by a light wave on a non-reflecting surface of 30 cm^2 area during 40 minutes of time span, the energy flux of light just before it falls on the surface is _____. W/cm^2 .

(Round off to the Nearest Integer)

(Assume complete absorption and normal incidence conditions are there)

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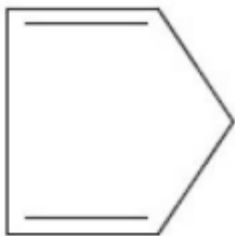
(Assume complete absorption and normal incidence conditions are there)

Solution:

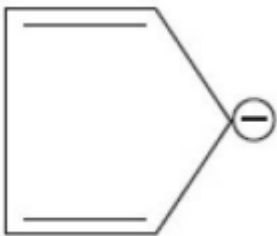
$$F = \frac{IA}{C}$$
$$I = \frac{FC}{A} = \frac{2.5 \times 10^{-6} \times 3 \times 10^8}{30} = 25 \text{ W/cm}^2$$

Q31: Which of the following is an aromatic compound?

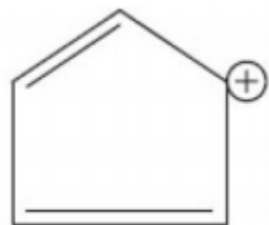
(A)



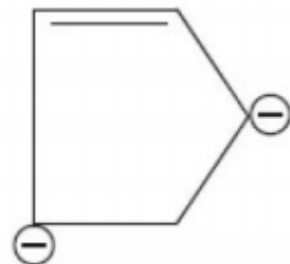
(B)



(C)

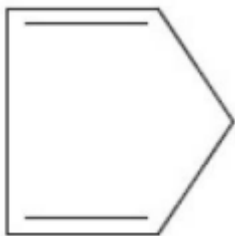


(D)

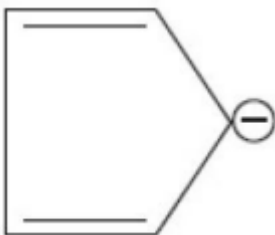


Q31: Which of the following is an aromatic compound?

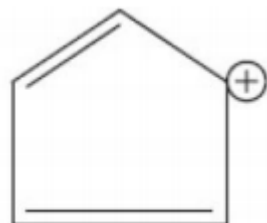
(A)



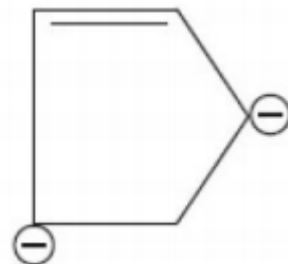
(B)



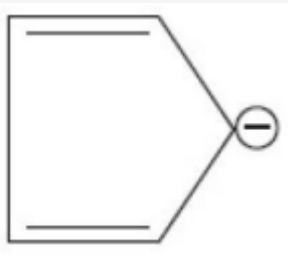
(C)



(D)



Solution:



6 π electron system
It is aromatic in nature.

Q32: The incorrect statement(s) about heavy water is (are)

- (A) used as a moderator in nuclear reactor
- (B) obtained as a by-product in fertilizer industry
- (C) used for the study of reaction mechanism
- (D) has a higher dielectric constant than water

(A) (B) only

(B) (D) only

(C) (C) only

(D) (B) and (D) only

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(C) (C) only

(D) (B) and (D) only

Solution:

D_2O = 78.06 (Dielectric constant)

H_2O = 78.39 (Dielectric constant)

Q33: Given below are two statements:

Statement I: Potassium permanganate on heating at 573 K forms potassium manganate.

Statement II: Both potassium permanganate and potassium manganate are tetrahedral and paramagnetic in nature.

In the light of the above statements, choose the most appropriate answer from the options given below:

- (A) Both statement I and statement II are false
- (B) Statement I is true but statement II is false
- (C) Statement I is false but statement II is true
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Q33: Given below are two statements:

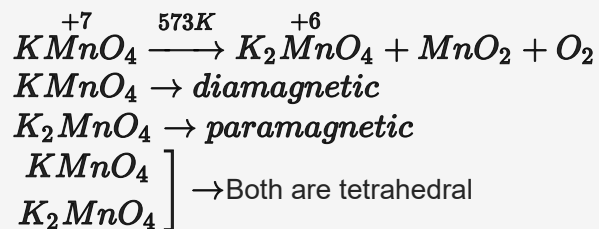
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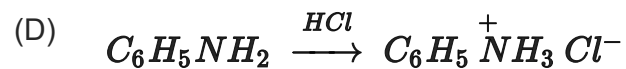
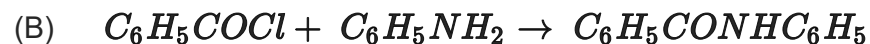
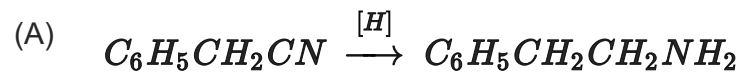
In the light of the above statements, choose the most appropriate answer from the options given below:

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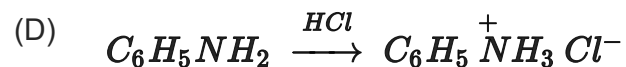
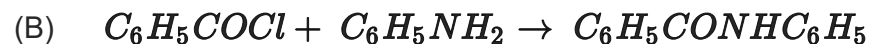
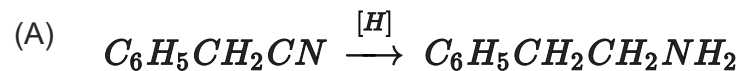
Solution:



Q34: Which of the following reaction is an example of ammonolysis?



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Solution:



Q35: With respect to drug-enzyme interaction, identify the wrong statement.

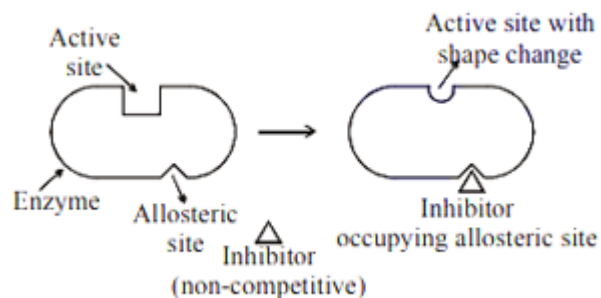
- (A) Competitive inhibitor binds to the enzyme's active site
- (B) Allosteric inhibitor competes with the enzyme's active site
- (C) Allosteric inhibitor changes the enzyme's active site
- (D) Non-Competitive inhibitor binds to the allosteric site

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Solution:

Some drug do not bind to the Enzyme's active site. These bind to a different site of enzyme which called allosteric site. This binding of inhibitor at allosteric site changes the shape of the active site in such a way that substrate cannot recognize it. Such inhibitor is known as Non-competitive inhibitor.



Q36: Given below are two statements:

Statement I: Retardation factor (R_f) can be measured in meter/centimetre.

Statement II: R_f value of a compound remains constant in all solvents.

Choose the most appropriate answer from the options given below :

- (A) Statement I is true but statement II is false
- (B) Statement I is false but statement II is true
- (C) Both statement I and statement II are true
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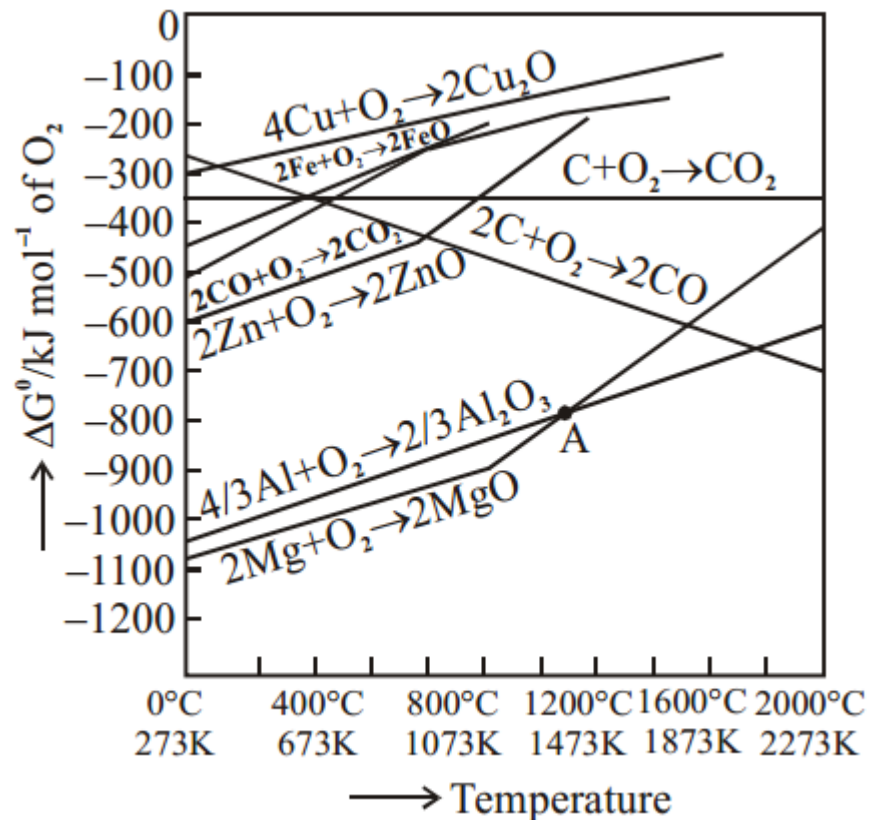
Choose the most appropriate answer from the options given below :

- (A) Statement I is true but statement II is false
- (B) Statement I is false but statement II is true
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- (D) Both statement I and statement II are false

Solution:

R_f (Retardation factor is dimension less) and lengths should be measured in the same units.
 R_f value changes with the solvent due to different degree of adsorption.

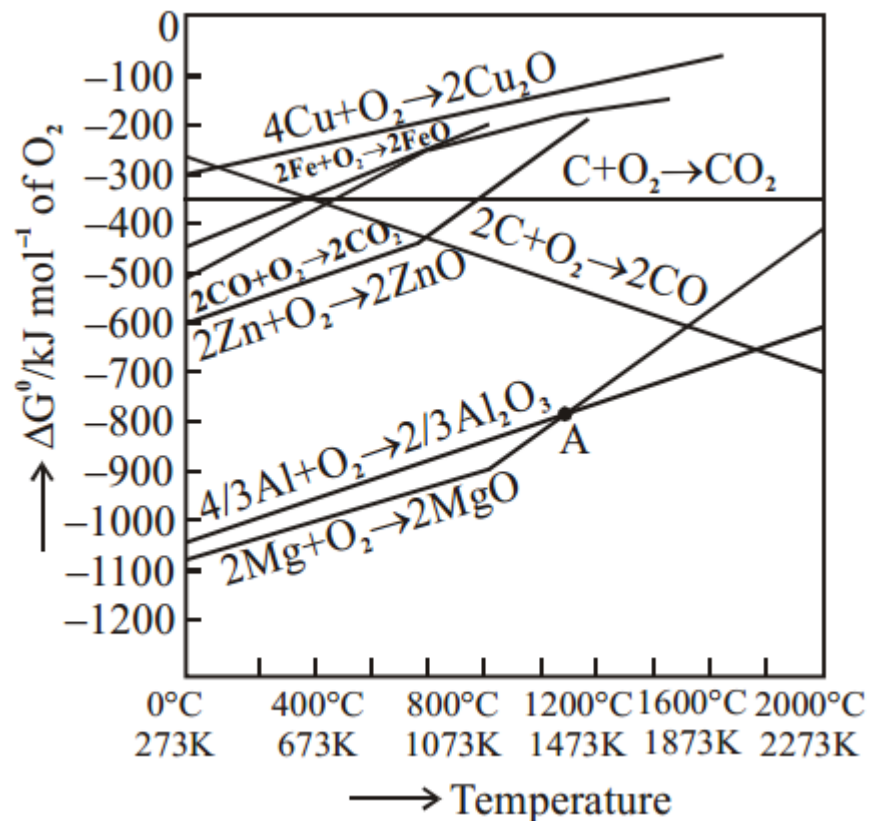
Q37: The point of intersection and sudden increase in the slope, in the diagram given below, respectively, indicates:



- (A) $\Delta G = 0$ and reduction of the metal oxide
- (B) $\Delta G > 0$ and decomposition of the metal oxide
- (C) $\Delta G < 0$ and decomposition of the metal oxide

(D) $\Delta G = 0$ and melting or boiling point of the metal oxide

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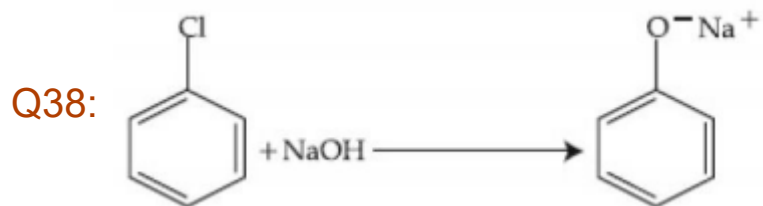
(D)

$\Delta G = 0$ and melting or boiling point of the metal oxide

Solution:

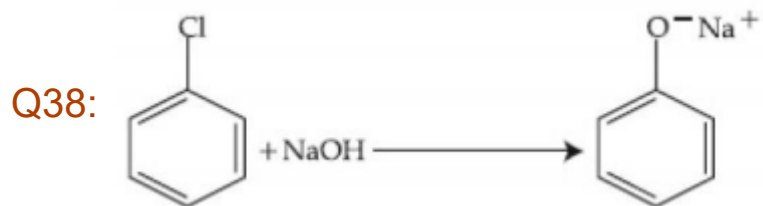
At the point of intersection $\Delta G = 0$ for involved reaction.

The sudden increase in the slope due to melting or boiling point of the metal oxide



The above reaction requires which of the following reaction conditions?

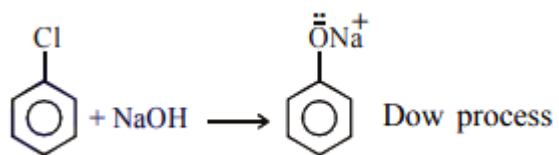
- (A) 623 K, Cu, 300 atm
- (B) 573 K, 300 atm
- (C) 573 K, Cu, 300 atm
- (D) 623 K, 300 atm



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- (A) 623 K, Cu, 300 atm
- (B) 573 K, 300 atm
- (C) 573 K, Cu, 300 atm
- (D) 623 K, 300 atm

Solution:



Temperature = 623 K
Pressure = 300 atm

Q39: What is the spin-only magnetic moment value (BM) of a divalent metal ion with atomic number 25, in its aqueous solution?

- (A) 5.26
- (B) 5.92
- (C) zero
- (D) 5.0

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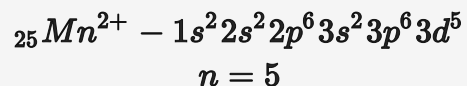
(A) 5.26

(B) 5.92

(C) zero

(D) 5.0

Solution:



$$\begin{aligned}\text{Spin - only magnetic moment} &= \sqrt{n(n+2)} \text{ BM} \\ &= \sqrt{5(5+2)} = \sqrt{35} = 5.92 \text{ BM}\end{aligned}$$

Q40: A central atom in a molecule has two lone pairs of electrons and forms three single bonds. The shape of this molecule is :

- (A) see-saw
- (B) T-shaped
- (C) planar triangular
- (D) trigonal pyramidal

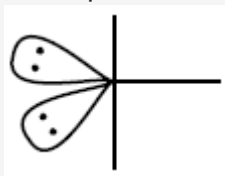
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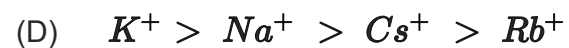
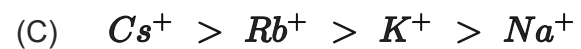
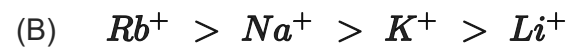
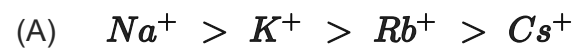
Solution:

$$2 \text{ L.P} + 3 \text{ B.P} = 5 \text{ VSEP } (sp^3d)$$

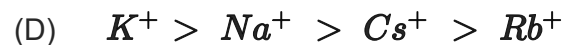
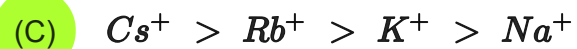
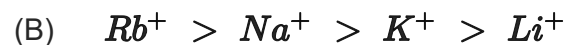
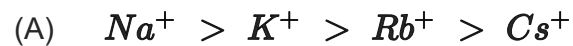
T- Shape



Q41: The correct order of conductivity of ions in water is :



Q41: The correct order of conductivity of ions in water is :



Solution:

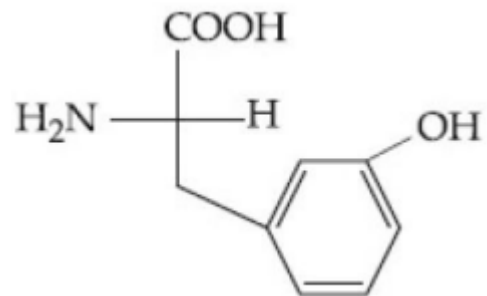
Cs_{aq}^+ has lower hydrated radius so its electrical conductivity is higher.

Order will be:

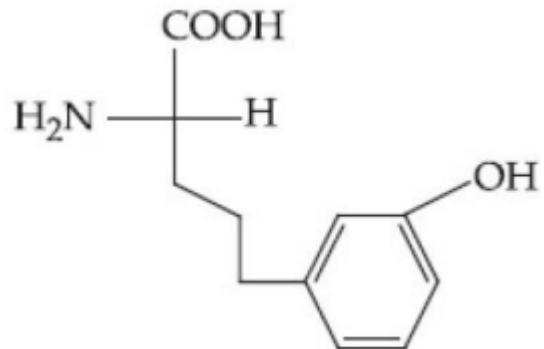


Q42: Which of the following is correct structure of tyrosine?

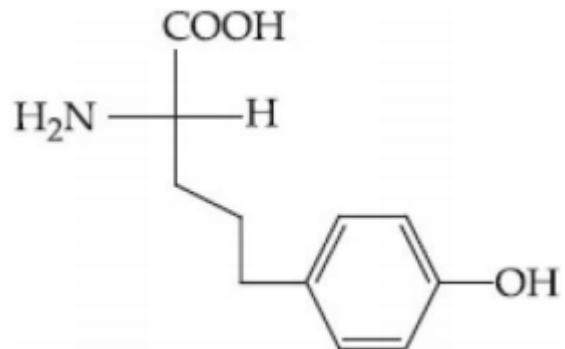
(A)



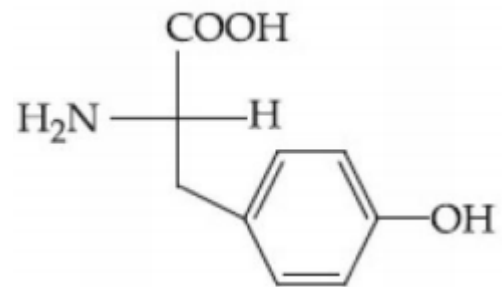
(B)



(C)

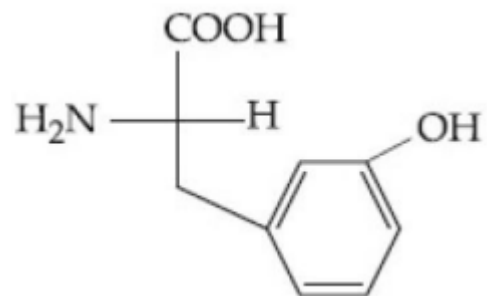


(D)

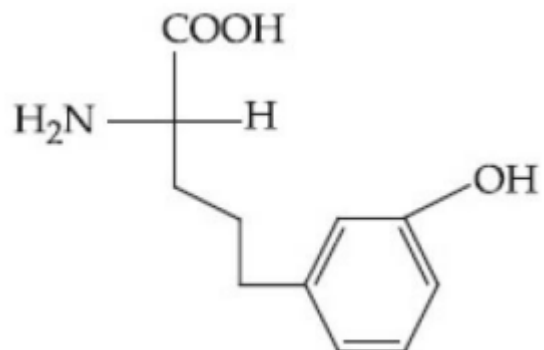


Q42: Which of the following is correct structure of tyrosine?

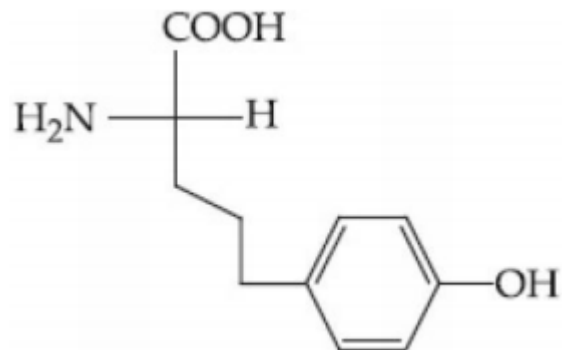
(A)



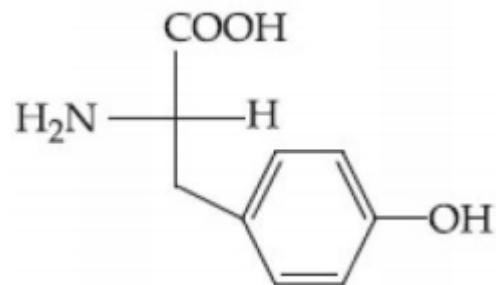
(B)



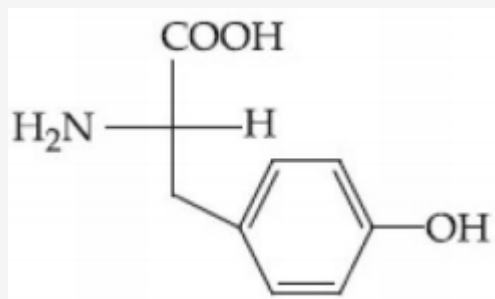
(C)



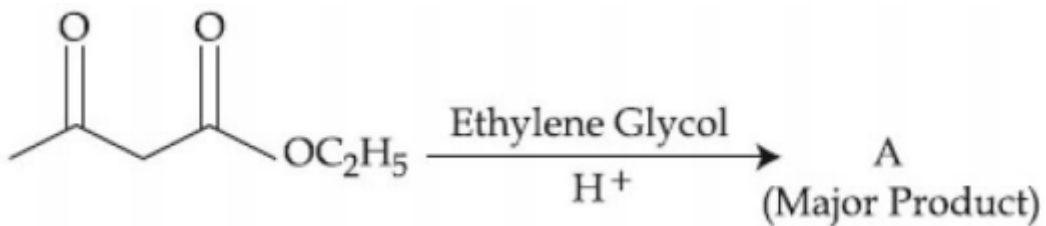
(D)



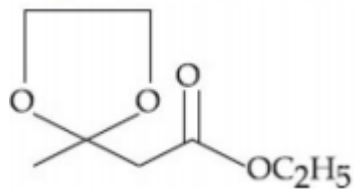
Solution:



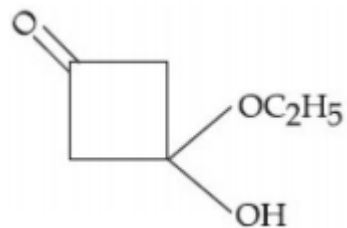
Q43:



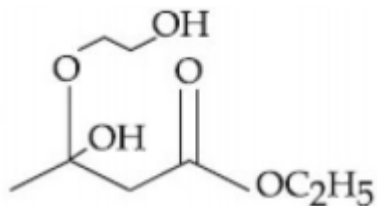
(A)



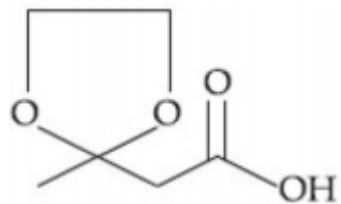
(B)



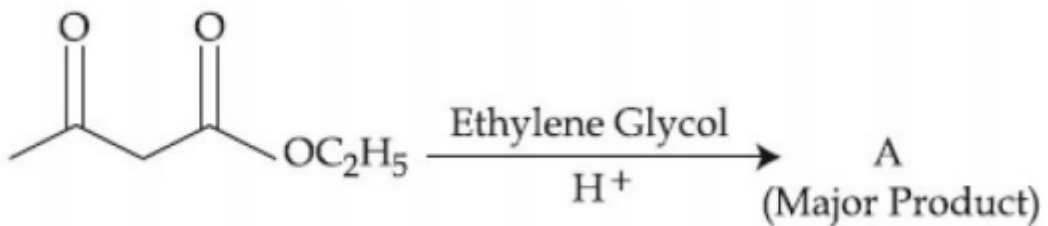
(C)



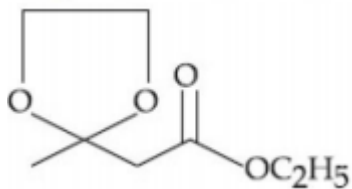
(D)



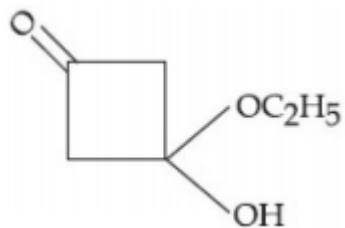
Q43:



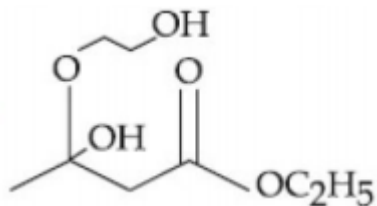
(A)



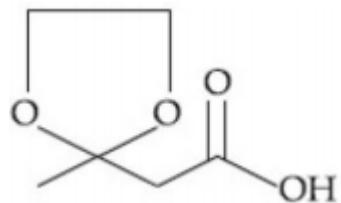
(B)



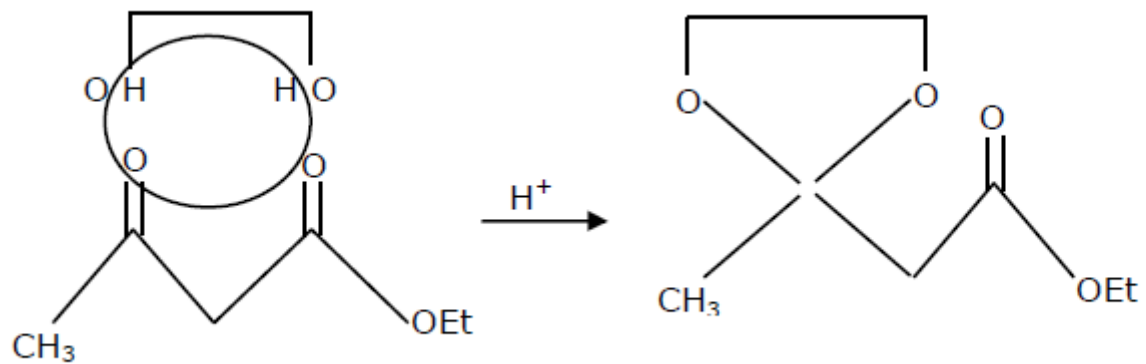
(C)



(D)



Solution:



Q44: The absolute value of the electron gain enthalpy of halogens satisfies:

- (A) $\text{Cl} > \text{F} > \text{Br} > \text{I}$
- (B) $\text{I} > \text{Br} > \text{Cl} > \text{F}$
- (C) $\text{Cl} > \text{Br} > \text{F} > \text{I}$
- (D) $\text{F} > \text{Cl} > \text{Br} > \text{I}$

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(B) $\text{I} > \text{Br} > \text{Cl} > \text{F}$

(C) $\text{Cl} > \text{Br} > \text{F} > \text{I}$

(D) $\text{F} > \text{Cl} > \text{Br} > \text{I}$

Solution:

Chlorine has higher electron gain enthalpy than fluorine due to less electron density. Down the group in halogen electron gain enthalpy decreases.

Q45: Which of the following compound cannot act as a Lewis base?



Q45: Which of the following compound cannot act as a Lewis base?

(A) SF_4

(B) PCl_5

(C) ClF_3

(D) NF_3

Solution:

PCl_5 do not have lone pair of electron on the central atom for donation so it cannot act as lewis base.

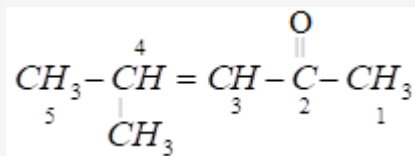
Q46: Mesityl oxide is a common name of :

- (A) 4-Methyl pent-3-en-2-one
- (B) 2,4-Dimethyl pentan-3-one
- (C) 3-Methyl cyclohexane carbaldehyde
- (D) 2-Methyl cyclohexanone

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- (D) 2-Methyl cyclohexanone

Solution:



(Mesityl oxide)

4-methyl pent – 3 – en – 2 – one

Q47: Reducing smog is a mixture of:

- (A) Smoke, fog and $CH_2 = CH - CHO$
- (B) Smoke, fog and O_3
- (C) Smoke, fog and SO_2
- (D) Smoke, fog and N_2O_3

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- (B) Smoke, fog and O_3
- (C) Smoke, fog and SO_2
- (D) Smoke, fog and N_2O_3

Solution:

Reducing smog = smoke + fog + SO_2

Q48: A colloidal system consisting of a gas dispersed in a solid is called a/an :

- (A) aerosol
- (B) gel
- (C) foam
- (D) solid sol

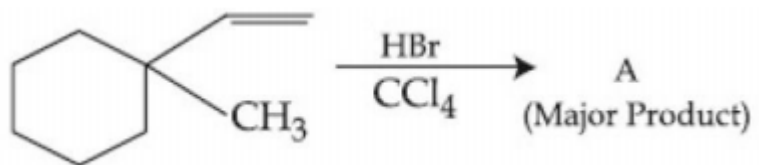
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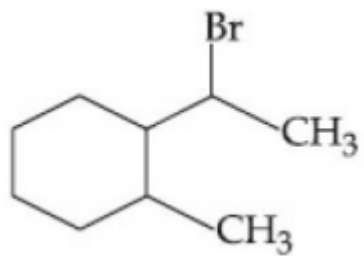
Solution:

Dispersed phase	Dispersion medium	Type of colloid
Gas	Solid	solid sol

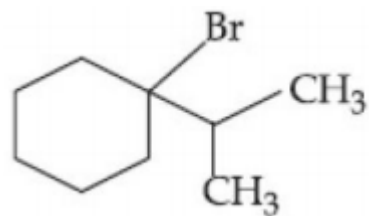
Q49:



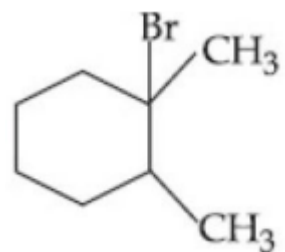
(A)



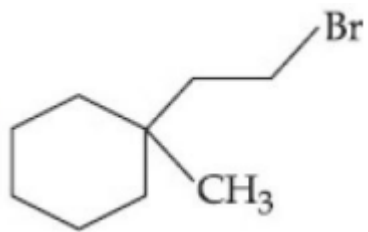
(B)

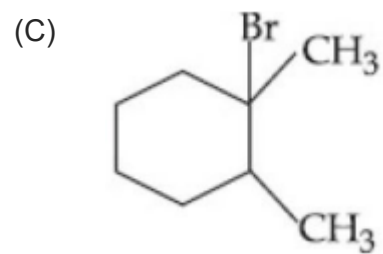
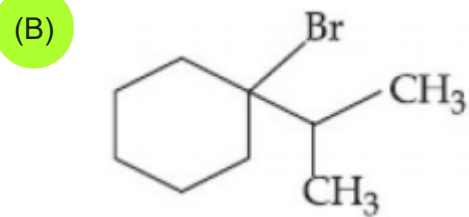
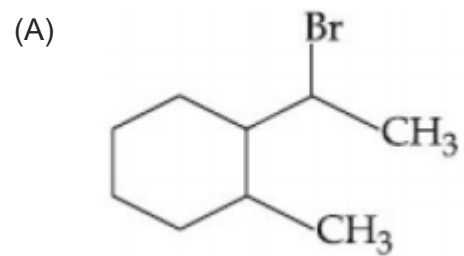
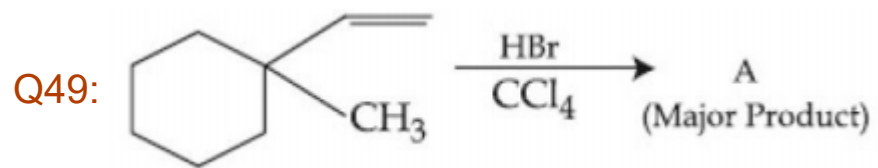


(C)

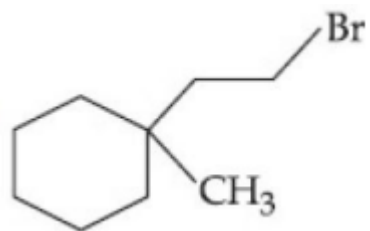


(D)

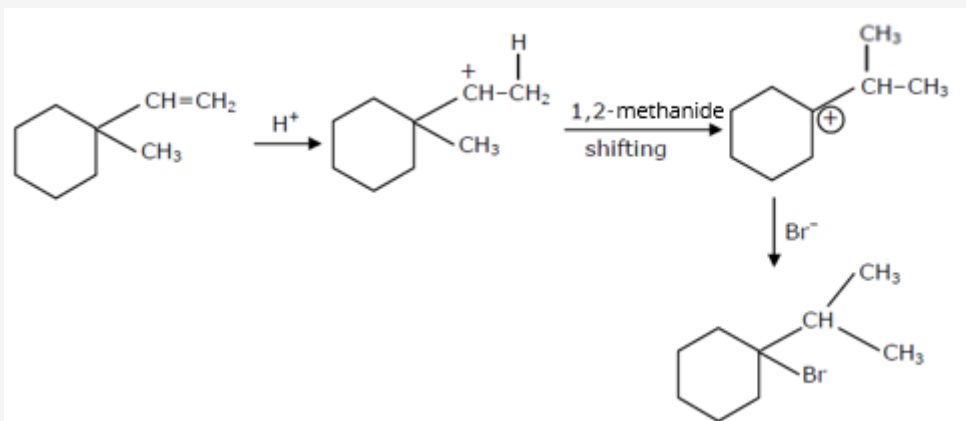




(D)



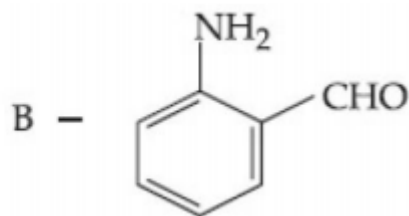
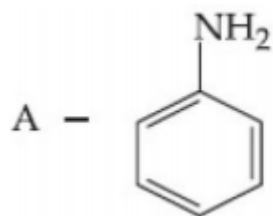
Solution:



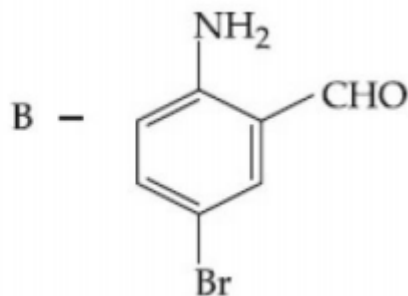
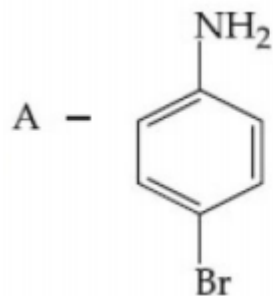
Q50: Hoffmann bromomide degradation of benzamide gives product A, which upon heating with CHCl_3 and NaOH gives product B.

The structures of A and B are:

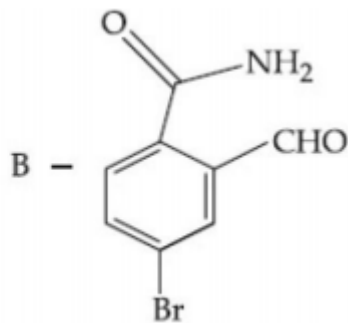
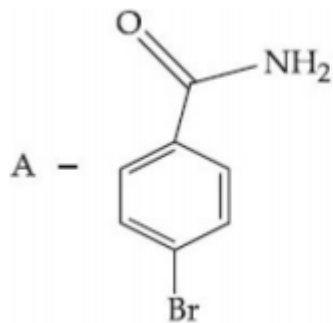
(A)



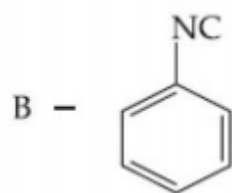
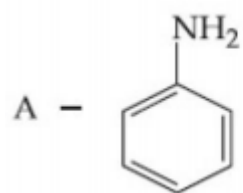
(B)



(C)



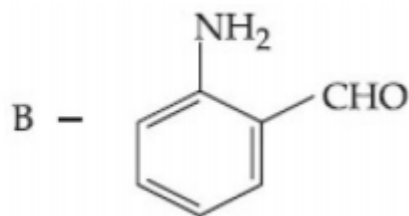
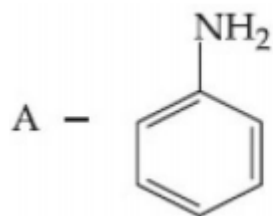
(D)



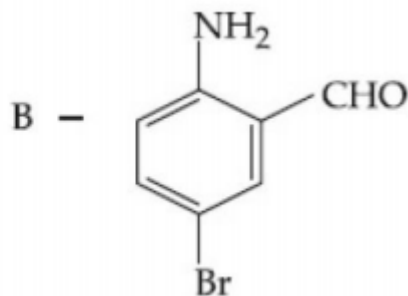
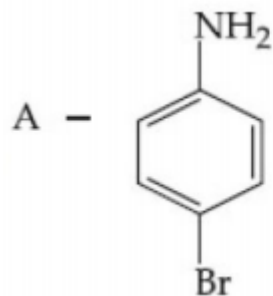
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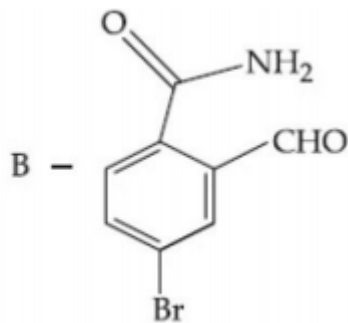
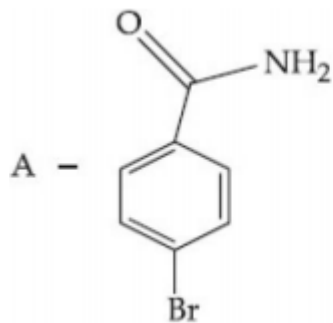
(A)



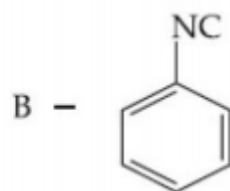
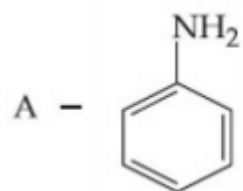
(B)



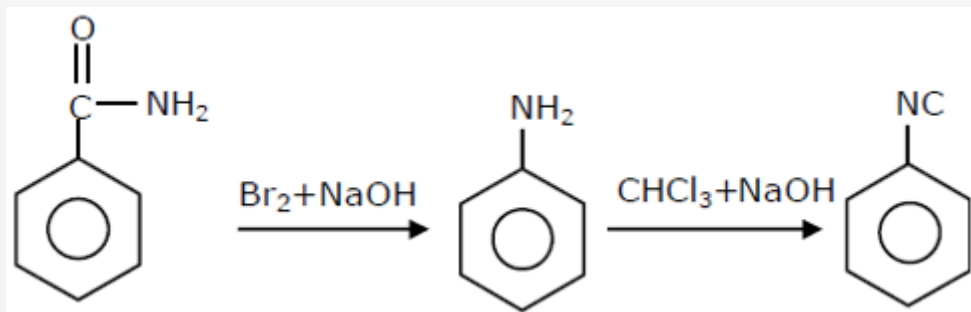
(C)



(D)



Solution:



Q51: The pressure exerted by a non-reactive gaseous mixture of 6.4 g of methane and 8.8 g of carbon dioxide in a 10 L vessel at 27°C is kPa.

(Round off to the Nearest Integer).

[Assume gases are ideal, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Atomic masses : C : 12.0 u, H : 1.0 u, O : 16.0 u]

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[Assume gases are ideal, $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$

Atomic masses : C : 12.0 u, H : 1.0 u, O : 16.0 u]

Solution:

$$V = 10L, T = 27^\circ C = 300K$$

$$(m)_{\text{methane}} = 6.4g, (m)_{\text{co}_2} = 8.8g$$

$$PV = n_{\text{total}}RT$$

$$P \times 10 \times 10^{-3} = \left(\frac{6.4}{16} + \frac{8.8}{44} \right) \times 8.314 \times 300$$

$$P \times 10^{-2} = (0.4 + 0.2) \times 8.314 \times 300$$

$$P = 149652 \text{ Pa}$$

$$P = 149.652 \text{ kPa} \approx 150 \text{ kPa}$$

Q52: The standard enthalpies of formation of Al_2O_3 and CaO are $-1675 \text{ kJ mol}^{-1}$ and -635 kJ mol^{-1} respectively.

For the reaction

$3\text{CaO} + 2\text{Al} \rightarrow 3\text{Ca} + \text{Al}_2\text{O}_3$ the standard reaction enthalpy $\Delta_r H^0 = \dots\dots\dots$ kJ.

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Solution:

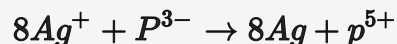
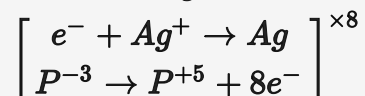
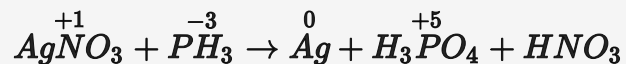
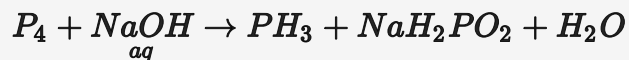
$$\begin{aligned}\Delta H_f^0 &= \Delta H_f^0 (\text{Products}) - \Delta H_f^0 (\text{Reactants}) \\ &= \Delta H_f^0 (\text{Al}_2\text{O}_3) - 3 \times \Delta H_f^0 (\text{CaO}) \\ &= -1675 - 3(-635) \\ &= 230 \text{ kJ}\end{aligned}$$

Q53: The reaction of white phosphorus on boiling with alkali in inert atmosphere resulted in the formation of product 'A'. The reaction of 1 mol of 'A' with excess of AgNO_3 in aqueous medium gives mol(s) of Ag. (Round off to the Nearest Integer).

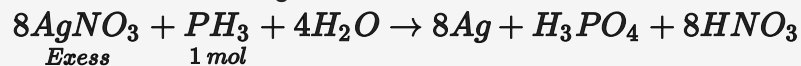
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8.00

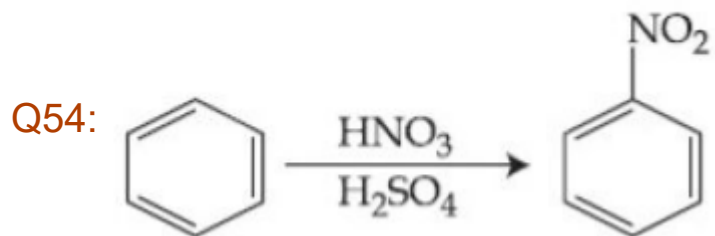
Solution:



So final reaction along with stoichiometric coeff. is.

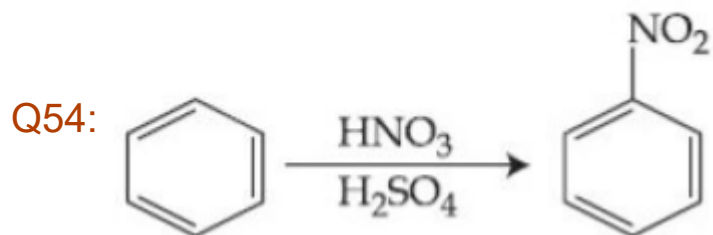


Hence 1 mol produce 8 mol Ag



In the above reaction, 3.9 g of benzene on nitration gives 4.92 g of nitrobenzene. The percentage yield of nitrobenzene in the above reaction is %. (Round off to the Nearest Integer).

(Given atomic mass: C : 12.0 u, H : 1.0 u, O : 16.0 u, N : 14.0 u)



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(Given atomic mass: C : 12.0 u, H : 1.0 u, O : 16.0 u, N : 14.0 u)

Solution:

$$\text{Moles of } C_6H_6 = \frac{3.9}{78} = 0.05$$

$$\text{Moles of } C_6H_5NO_2 = \frac{4.92}{123} = 0.04$$

By conserving moles of carbon, mole of $C_6H_5NO_2$

Formed theoretically are 0.05

$$\Rightarrow \% \text{ yield} = \frac{\text{moles formed actually}}{\text{moles formed theoretically}} \times 100$$

$$\Rightarrow \% \text{ yield} = \frac{0.04}{0.05} \times 100 = 80\%$$

Q55: 0.01 moles of a weak acid HA ($K_a = 2.0 \times 10^{-6}$) is dissolved in 1.0 L of 0.1 M HCl solution.

The degree of dissociation of HA is $\times 10^{-5}$ (Round off to the Nearest Integer).

[Neglect volume change on adding HA.

Assume degree of dissociation $\ll 1$]

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[Neglect volume change on adding HA.

Assume degree of dissociation $\ll 1$]

2.00

Solution:

$$\begin{array}{ccccc}
 HA & \rightleftharpoons & H^+ & + & A^- \\
 C_1 & 0.01 & 0 & & 0 \\
 C_{eq} & 0.01(1-\alpha) & 0.01\alpha + 0.1 & & 0.01\alpha \\
 & \simeq 0.01 & \simeq 0.1 & &
 \end{array}$$

$$K_a = \frac{[H^+][A^-]}{[HA]}$$

$$2 \times 10^{-6} = \frac{(0.1)(0.01\alpha)}{0.01}$$

$$\alpha = 2 \times 10^{-5}$$

Q56: For a certain first-order reaction 32% of the reactant is left after 570 s. The rate constant of this reaction is $\times 10^{-3} \text{ s}^{-1}$. [Given: $\log_{10} 2 = 0.301$, $\ln 10 = 2.303$]

Q56: For a certain first-order reaction 32% of the reactant is left after 570 s. The rate constant of this reaction is $\times 10^{-3} \text{ s}^{-1}$. [Given: $\log_{10} 2 = 0.301$, $\ln 10 = 2.303$]

2.02

Solution:

$$k = \frac{1}{t} \ln \left[\frac{a}{a-x} \right]$$

$$k = \frac{2.303}{570} \log \left(\frac{100}{32} \right)$$

$$k = \frac{2.303}{570} [\log(10^2) - \log 2^5]$$

$$k = \frac{2.303}{570} \times 0.5$$

$$k = 2.02 \times 10^{-3} \text{ s}^{-1}$$

Q57: The oxygen dissolved in water exerts a partial pressure of 20 kPa in the vapour above water. The molar solubility of oxygen in water is $\times 10^{-5}$ mol dm⁻³.

(Round off to the Nearest Integer).

[Given: Henry's law constant = $K_H = 8.0 \times 10^4$ kPa for O₂.

Density of water with dissolved oxygen = 1.0 kg dm⁻³]

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(Round off to the Nearest Integer).

[Given: Henry's law constant = $K_H = 8.0 \times 10^4$ kPa for O₂.

Density of water with dissolved oxygen = 1.0 kg dm⁻³]

Solution:

$$P_{(g)} = [K_H] \chi$$

$$20 \times 10^3 = [8.0 \times 10^4 \times 10^3] \times \text{Solubility}$$

$$\text{Solubility} = \frac{20 \times 10^3}{8.0 \times 10^7} = 2.5 \times 10^{-4}$$

$$\text{Solubility} = 25 \times 10^{-5}$$

Q58: The mole fraction of a solute in a 100 molal aqueous solution is $\times 10^{-2}$.
(Round off to the Nearest Integer).

[Given: Atomic masse: H : 1.0 u, O : 16.0 u]

Q58: The mole fraction of a solute in a 100 molal aqueous solution is $\times 10^{-2}$.
(Round off to the Nearest Integer).

[Given: Atomic masse: H : 1.0 u, O : 16.0 u]

64.00

Solution:

Let weight of $H_2O = 1000g$

Moles of solute = 100

(mole) $H_2O = \frac{1000}{18}$

Mole fraction of solute = $\frac{\text{mole of solute}}{\text{Total moles}}$

$$= \frac{100}{100 + \frac{1000}{18}} = \frac{1800}{2800}$$

$$X_{\text{solute}} = 64 \times 10^{-2}$$

Q59: 15 mL of aqueous solution of Fe^{2+} in acidic medium completely reacted with 20 mL of 0.03 M aqueous $Cr_2O_7^{2-}$. The molarity of the Fe^{2+} solution is $\times 10^{-2}$ M. (Round off to the Nearest Integer).

Q59: 15 mL of aqueous solution of Fe^{2+} in acidic medium completely reacted with 20 mL of 0.03 M aqueous $Cr_2O_7^{2-}$. The molarity of the Fe^{2+} solution is $\times 10^{-2}$ M. (Round off to the Nearest Integer).

24.00

Solution:

By law of equivalence Meq of Fe^{2+} = Meq of $Cr_2O_7^{2-}$

$$M \times 15 \times 1 = 0.03 \times 6 \times 20$$

$$M = 0.24 \text{ M} = 24 \times 10^{-2} \text{ M}$$

Q60: A certain orbital has $n = 4$ and $m_l = -3$. The number of radial nodes in this orbital is (Round off to the Nearest Integer).

Q60: A certain orbital has $n = 4$ and $m_l = -3$. The number of radial nodes in this orbital is (Round off to the Nearest Integer).

Solution:

$$\text{Number of radial nodes} = n - \ell - 1$$

$$n = 4, m_l = -3 \text{ so } \ell = 3$$

$$\text{Radial nodes} = 4 - 3 - 1 = 0$$

Q61: The line $2x - y + 1 = 0$ is a tangent to the circle at the point $(2, 5)$ and the centre of the circle lies on $x - 2y = 4$. Then, the radius of the circle is :

(A) $3\sqrt{5}$

(B) $5\sqrt{3}$

(C) $5\sqrt{4}$

(D) $4\sqrt{5}$

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(A) $3\sqrt{5}$

(B) $5\sqrt{3}$

(C) $5\sqrt{4}$

(D) $4\sqrt{5}$

Solution:

Equation of normal passing through $(2, 5)$ is $x + 2y = 12$

Let centre be (h, k)

$$\therefore h - 2k = 4$$

$$h + 2k = 12$$

$$h = 8, k = 2$$

$$\therefore \text{Radius} = \sqrt{36 + 9} = \sqrt{45} = 3\sqrt{5}$$

Q62: Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 7\hat{i} + \hat{j} - 6\hat{k}$

If $\vec{r} \times \vec{a} = \vec{r} \times \vec{b}$, $\vec{r} \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -3$, then $\vec{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k})$ is

equal to :

(A) 12

(B) 13

(C) 10

(D) 8

Q62: Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 7\hat{i} + \hat{j} - 6\hat{k}$

If $\vec{r} \times \vec{a} = \vec{r} \times \vec{b}$, $\vec{r} \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -3$, then $\vec{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k})$ is

equal to :

(A) 12

(B) 13

(C) 10

(D) 8

Solution:

$$\vec{r} \times \vec{a} - \vec{r} \times \vec{b} = \vec{0}$$

$$\vec{r} \times (\vec{a} - \vec{b}) = \vec{0}$$

$$\Rightarrow \vec{r} = \lambda (\vec{a} - \vec{b})$$

$$\Rightarrow \vec{r} = \lambda (-5\hat{i} - 4\hat{j} + 10\hat{k})$$

$$\vec{r} \cdot \vec{c} = -3 \Rightarrow \lambda (-5 - 8 + 10) = -3$$

$$\Rightarrow \lambda = 1$$

$$\therefore \vec{r} = -5\hat{i} - 4\hat{j} + 10\hat{k}$$

Q63: In a triangle PQR , the co-ordinates of the points P and Q are $(-2, 4)$ and $(4, -2)$ respectively. If the equation of the perpendicular bisector of PR is $2x - y + 2 = 0$, then the centre of the circumcircle of the ΔPQR is :

- (A) $(0, 2)$
- (B) $(-2, -2)$
- (C) $(1, 4)$
- (D) $(-1, 0)$

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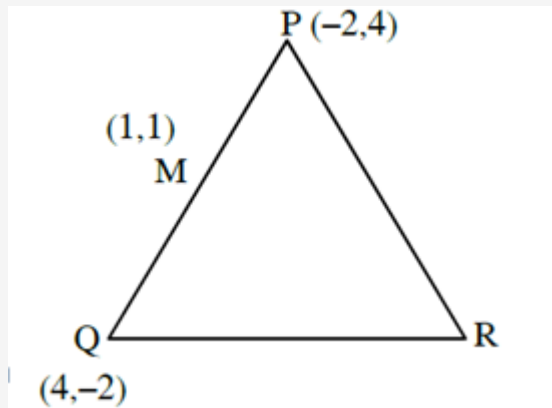
(A) $(0, 2)$

(B) $(-2, -2)$

(C) $(1, 4)$

(D) $(-1, 0)$

Solution:



Perpendicular bisector of PR $2x - y + 2 = 0$... (i)

Mid point of $PQ \rightarrow M(1, 1)$

equation of perpendicular bisector of $PQ : x - y = 0$ (ii)

So, Point of intersection of equation (i) & (ii) is circumcentre

So, Circumcentre $(-2, -2)$

Q64: Two dices are rolled. If both dices have six faces numbered 1, 2, 3, 5, 7 and 11, then the probability that the sum of the numbers on the top faces is less than or equal to 8 is :

(A) $\frac{5}{12}$

(B) $\frac{1}{2}$

(C) $\frac{17}{36}$

(D) $\frac{4}{9}$

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(A) $\frac{5}{12}$

(B) $\frac{1}{2}$

(C) $\frac{17}{36}$

(D) $\frac{4}{9}$

Solution:

$$n(S) = 36$$

Possible ordered pair;

$(1, 1), (1, 2), (1, 3), (1, 5), (1, 7), (2, 1), (2, 2), (2, 3), (2, 5), (3, 1), (3, 2), (3, 3), (3, 5), (5, 1), (5, 2), (5, 3), (7, 1)$

Number of ordered pairs = 17

$$\text{Probability} = \frac{17}{36}$$

Q65: The area of the triangle with vertices $A(z)$, $B(iz)$ and $C(z + iz)$ is :

(A) $\frac{1}{2} |z + iz|^2$

(B) $\frac{1}{2}$

(C) $\frac{1}{2} |z|^2$

(D) 1

Q65: The area of the triangle with vertices $A(z)$, $B(iz)$ and $C(z + iz)$ is :

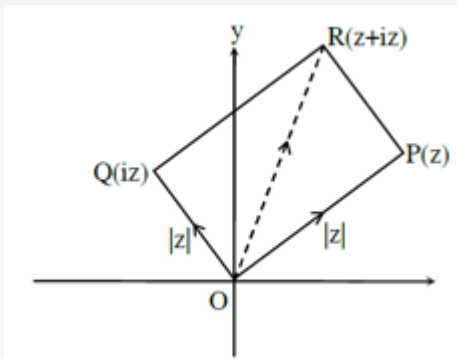
(A) $\frac{1}{2} |z + iz|^2$

(B) $\frac{1}{2}$

(C) $\frac{1}{2} |z|^2$

(D) 1

Solution:



$$\begin{aligned}\text{Area of } \Delta &= \frac{1}{2} (\text{area of square}) \\ &= \frac{1}{2} |z|^2\end{aligned}$$

Q66: The system of equations $kx + y + z = 1$, $x + ky + z = k$ and $x + y + zk = k^2$ has no solution is k is equal to :

- (A) 1
- (B) -2
- (C) -1
- (D) 0

Q66: The system of equations $kx + y + z = 1$, $x + ky + z = k$ and $x + y + zk = k^2$ has no solution is k is equal to :

(A) 1

(B) -2

(C) -1

(D) 0

Solution:

$$D = \begin{vmatrix} k & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{vmatrix} = 0$$

$$k(k^2 - 1) - (k - 1) + (1 - k) = 0$$

$$(k - 1)(k^2 + k - 1 - 1) = 0$$

$$(k - 1)(k - 1)(k + 2) = 0$$

$$k = 1, k = -2$$

For $k = 1$ equation becomes identical so $k = -2$ for no solution

Q67: Which of the following is true for $y(x)$ that satisfies the differential equation

$$\frac{dy}{dx} = xy - 1 + x - y; y(0) = 0 :$$

(A) $y(1) = e^{\frac{1}{2}} - e^{-\frac{1}{2}}$

(B) $y(1) = 1$

(C) $y(1) = e^{\frac{1}{2}} - 1$

(D) $y(1) = e^{-\frac{1}{2}} - 1$

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(B) $y(1) = 1$

(C) $y(1) = e^{\frac{1}{2}} - 1$

(D) $y(1) = e^{-\frac{1}{2}} - 1$

Solution:

$$\frac{dy}{dx} = (x-1)y + (x-1)$$

$$\frac{dy}{dx} = (x-1)(y+1)$$

$$\frac{dy}{y+1} = (x-1) dx$$

$$\ln(y+1) = \frac{x^2}{2} - x + c$$

$$x=0, y=0$$

$$\Rightarrow c=0$$

$$\therefore \ln(y+1) = \frac{x^2}{2} - x$$

$$\text{Putting } x=1, \ln(y+1) = \frac{1}{2} - 1 = -\frac{1}{2}$$

$$y+1 = e^{-\frac{1}{2}}$$

$$y = e^{-\frac{1}{2}} - 1$$

$$\therefore y(1) = e^{-\frac{1}{2}} - 1$$

Q68: If the Boolean expression $(p \rightarrow q) \leftrightarrow (q * (\sim p))$ is a tautology, then the Boolean expression $(p * (\sim q))$ is equivalent to :

- (A) $p \rightarrow \sim q$
- (B) $p \rightarrow q$
- (C) $q \rightarrow p$
- (D) $\sim q \rightarrow p$

Q68: If the Boolean expression $(p \rightarrow q) \leftrightarrow (q * (\sim p))$ is a tautology, then the Boolean expression $(p * (\sim q))$ is equivalent to :

(A) $p \rightarrow \sim q$

(B) $p \rightarrow q$

(C) $q \rightarrow p$

(D) $\sim q \rightarrow p$

Solution:

As the expression $(p \rightarrow q) \leftrightarrow (q * (\sim p))$ is a tautology, so

$$q * (\sim p) \equiv (p \rightarrow q)$$

Interchanging p and q , we get

$$p * (\sim q) \equiv (q \rightarrow p)$$

Q69: The value of $4 + \frac{1}{5 + \frac{1}{4 + \frac{1}{5 + \frac{1}{4 + \dots \infty}}}}$ is :

(A) $2 + \frac{4}{\sqrt{5}} \sqrt{30}$

(B) $5 + \frac{2}{5} \sqrt{30}$

(C) $4 + \frac{4}{\sqrt{5}} \sqrt{30}$

(D) $2 + \frac{2}{5} \sqrt{30}$

Q69: The value of $4 + \frac{1}{5 + \frac{1}{4 + \frac{1}{5 + \frac{1}{4 + \dots \infty}}}}$ is :

(A) $2 + \frac{4}{\sqrt{5}} \sqrt{30}$

(B) $5 + \frac{2}{5} \sqrt{30}$

(C) $4 + \frac{4}{\sqrt{5}} \sqrt{30}$

(D) $2 + \frac{2}{5} \sqrt{30}$

Solution:

$$y = 4 + \frac{1}{5 + \frac{1}{y}}$$

$$y = 4 + \frac{y}{5y+1}$$

$$5y^2 - 20y - 4 = 0$$

$$y = \frac{20 \pm \sqrt{400+80}}{10}$$

$$y = \frac{20 \pm 4\sqrt{30}}{10}, y > 0$$

$$y = \frac{10+2\sqrt{30}}{5}$$

Q70: The value of $\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(x - [x]^2) \cdot \sin^{-1}(x - [x]^2)}{x - x^3}$, where $[x]$ denotes the greatest integer

$\leq x$ is :

(A) $\frac{\pi}{4}$

(B) 0

(C) $\frac{\pi}{2}$

(D) π

Q70: The value of $\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(x - [x]^2) \cdot \sin^{-1}(x - [x]^2)}{x - x^3}$, where $[x]$ denotes the greatest integer

$\leq x$ is :

(A) $\frac{\pi}{4}$

(B) 0

(C) $\frac{\pi}{2}$

(D) π

Solution:

$$\lim_{x \rightarrow 0^+} \frac{\cos^{-1}(x - [x]^2) \sin^{-1}(x - [x]^2)}{x(1 - x^2)}$$
$$\Rightarrow \lim_{x \rightarrow 0^+} \frac{\cos^{-1} x \sin^{-1} x}{x} = \frac{\pi}{2}$$

Q71: If $A = \begin{pmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{pmatrix}$ and $\det \left(A^2 - \frac{1}{2} I \right) = 0$, then a possible value of α is :

(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{6}$

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(A) $\frac{\pi}{4}$

(B) $\frac{\pi}{2}$

(C) $\frac{\pi}{3}$

(D) $\frac{\pi}{6}$

Solution:

$$A^2 = \begin{bmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{bmatrix} \begin{bmatrix} 0 & \sin \alpha \\ \sin \alpha & 0 \end{bmatrix} = \begin{bmatrix} \sin^2 \alpha & 0 \\ 0 & \sin^2 \alpha \end{bmatrix}$$

$$A^2 - \frac{1}{2}I = \begin{bmatrix} \sin^2 \alpha & 0 \\ 0 & \sin^2 \alpha \end{bmatrix} - \begin{bmatrix} \frac{1}{2} & 0 \\ 0 & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} \sin^2 \alpha - \frac{1}{2} & 0 \\ 0 & \sin^2 \alpha - \frac{1}{2} \end{bmatrix}$$

$$\Rightarrow |A^2 - \frac{1}{2}I| = 0$$

$$\begin{vmatrix} \sin^2 \alpha - \frac{1}{2} & 0 \\ 0 & \sin^2 \alpha - \frac{1}{2} \end{vmatrix} = 0$$

$$\Rightarrow (\sin^2 \alpha - \frac{1}{2})^2 = 0$$

$$\Rightarrow \sin^2 \alpha = \frac{1}{2}$$

$$\Rightarrow \sin \alpha = \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}}$$

$$\alpha = \frac{\pi}{4}$$

Q72: If $\cot^{-1}(\alpha) = \cot^{-1}2 + \cot^{-1}8 + \cot^{-1}18 + \cot^{-1}32 + \dots$ upto 100 terms, then α is

- (A) 1.03
- (B) 1.02
- (C) 1.01
- (D) 1.00

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(A) 1.03

(B) 1.02

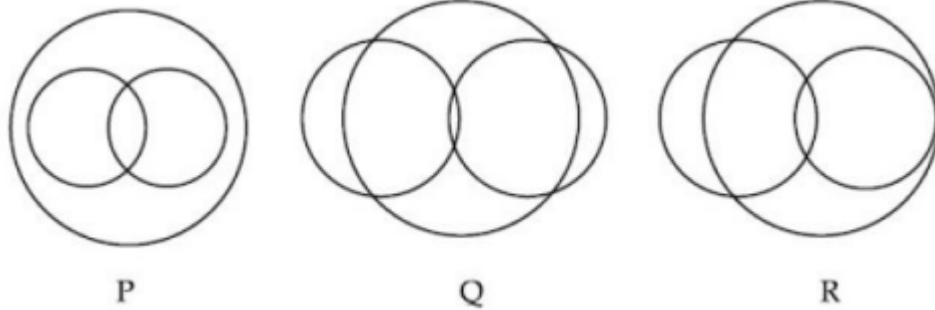
(C) 1.01

(D) 1.00

Solution:

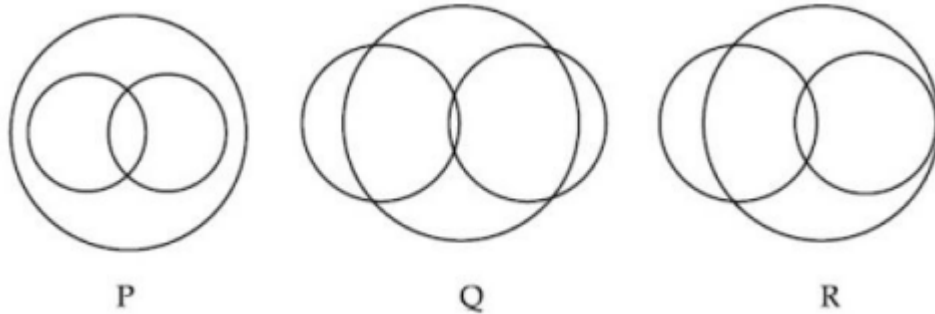
$$\begin{aligned} RHS &= \sum_{n=1}^{100} \cot^{-1} 2n^2 = \sum_{n=1}^{100} \tan^{-1} \left(\frac{2}{4n^2} \right) \\ &= \sum_{n=1}^{100} \tan^{-1} \left(\frac{(2n-1) - (2n+1)}{1 + (2n+1)(2n-1)} \right) \\ &= \sum_{n=1}^{100} \tan^{-1} (2n+1) - \tan^{-1} (2n-1) \\ &= \tan^{-1} 201 - \tan^{-1} 1 \\ &= \tan^{-1} \left(\frac{200}{202} \right) \\ &\Rightarrow \cot^{-1} \alpha = \cot^{-1} \left(\frac{101}{100} \right) \\ &\Rightarrow \alpha = 1.01 \end{aligned}$$

Q73: In a school, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagrams can justify the above statement ?



- (A) P and R
- (B) None of these
- (C) Q and R
- (D) P and Q

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- (A) P and R
- (B) None of these
- (C) Q and R
- (D) P and Q

Solution:

In all the three Venn diagrams, there is a region common to all the three circles, which is not true, as no student play all the three games. So none of them is correct.

Q74: The inverse of $y = 5^{\log x}$ is :

(A) $x = y^{\log 5}$

(B) $x = y^{\frac{1}{\log 5}}$

(C) $x = 5^{\log y}$

(D) $x = 5^{\frac{1}{\log y}}$

Q74: The inverse of $y = 5^{\log x}$ is :

(A) $x = y^{\log 5}$

(B) $x = y^{\frac{1}{\log 5}}$

(C) $x = 5^{\log y}$

(D) $x = 5^{\frac{1}{\log y}}$

Solution:

$$y = 5^{\log x}$$

$$y = x^{\log 5}$$

$$\text{So, } x = y^{\frac{1}{\log 5}}$$

Q75: The equation of the plane which contains the y-axis and passes through the point (1, 2, 3) is :

(A) $3x + z = 6$

(B) $x + 3z = 0$

(C) $x + 3z = 10$

(D) $3x - z = 0$

Q75: The equation of the plane which contains the y-axis and passes through the point (1, 2, 3) is :

(A) $3x + z = 6$

(B) $x + 3z = 0$

(C) $x + 3z = 10$

(D) $3x - z = 0$

Solution:

Let the equation of the plane be $a(x - 1) + b(y - 2) + c(z - 3) = 0$

y-axis lies on it D.R. 's of y-axis are 0, 1, 0

$$\therefore 0.a + 1.b + 0.c = 0$$

$$\Rightarrow b = 0$$

$\therefore$ Equation of plane is $a(x - 1) + c(z - 3) = 0$

$x = 0, z = 0$ also satisfy it

$$-a - 3c = 0 \Rightarrow a = -3c$$

$$-3c(x - 1) + c(z - 3) = 0$$

$$-3x + 3 + z - 3 = 0$$

$$3x - z = 0$$

Q76: If the fourth term in the expansion of $(x + x^{\log_2 x})^7$ is 4480, then the value of x where $x \in \mathbb{N}$ is equal to :

- (A) 4
- (B) 2
- (C) 3
- (D) 1

Q76: If the fourth term in the expansion of $(x + x^{\log_2 x})^7$ is 4480, then the value of x where $x \in \mathbb{N}$ is equal to :

(A) 4

(B) 2

(C) 3

(D) 1

Solution:

$${}^7C_3 x^4 (x^{\log_2 x})^3 = 4480$$

$$35x^4 (x^{\log_2 x})^3 = 4480$$

$$x^4 (x^{\log_2 x})^3 = 128$$

Take log w.r.t base 2

$$4\log_2 x + 3\log_2 (x^{\log_2 x}) = \log_2 128$$

$$\text{Let } \log_2 x = y$$

$$4y + 3y^2 = 7$$

$$\Rightarrow y = 1, \frac{-7}{3}$$

$$\Rightarrow \log_2 x = 1, \frac{-7}{3}$$

$$x = 2, x = 2^{\frac{-7}{3}}$$

Q77: Which of the following statements is incorrect for the function $g(\alpha)$ for $\alpha \in \mathbb{R}$ such

that $g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^\alpha x}{\cos^\alpha x + \sin^\alpha x} dx$

- (A) $g(\alpha)$ has an inflection point at $\alpha = -\frac{1}{2}$
- (B) $g(\alpha)$ is a strictly decreasing function
- (C) $g(\alpha)$ is a strictly increasing function
- (D) $g(\alpha)$ is an even function

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- (A) $g(\alpha)$ has an inflection point at $\alpha = -\frac{1}{2}$
- (B) $g(\alpha)$ is a strictly decreasing function
- (C) $g(\alpha)$ is a strictly increasing function
- (D) $g(\alpha)$ is an even function

Solution:

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^\alpha x}{\sin^\alpha x + \cos^\alpha x} dx \dots\dots\dots (i)$$

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\cos^\alpha x}{\sin^\alpha x + \cos^\alpha x} dx \dots\dots\dots (ii)$$

adding equation (i) and (ii)

$$2g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} 1 dx = \frac{\pi}{6}$$

$$\Rightarrow g(\alpha) = \frac{\pi}{12}$$

Q78: Choose the incorrect statement about the two circles whose equations are given below :

$$x^2 + y^2 - 10x - 10y + 41 = 0 \text{ and}$$

$$x^2 + y^2 - 16x - 10y + 80 = 0$$

- (A) Both circles pass through the centre of each other.
- (B) Both circles' centres lie inside region of one another.
- (C) Circles have two intersection points.
- (D) Distance between two centres is the average of radii of both the circles.

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- (A) Both circles pass through the centre of each other.
- (B) Both circles' centres lie inside region of one another.
- (C) Circles have two intersection points.
- (D) Distance between two centres is the average of radii of both the circles.

Solution:

$$C_1 (5, 5), C_2 (8, 5)$$

$$\text{Position of } C_1 (5, 5) \text{ in } S_2 = 0$$

$$= 25 + 25 - 80 - 50 + 80 = 0$$

$$\text{Position } C_2 (8, 5) \text{ in } S_1 = 0$$

$$\equiv 64 + 25 - 80 - 50 + 41 = 0$$

$\Rightarrow S_1$ and S_2 intersect each other and pass through center of each other

Q79: Team 'A' consists of 7 boys and n girls and Team 'B' has 4 boys and 6 girls. If a total of 52 single matches can be arranged between these two teams when a boy plays against a boy and a girl plays against a girl, then n is equal to :

- (A) 4
- (B) 5
- (C) 2
- (D) 6

Q79: Team 'A' consists of 7 boys and n girls and Team 'B' has 4 boys and 6 girls. If a total of 52 single matches can be arranged between these two teams when a boy plays against a boy and a girl plays against a girl, then n is equal to :

(A) 4

(B) 5

(C) 2

(D) 6

Solution:

$$7 \times 4 + 6 \times n = 52$$

$$6n = 24$$

$$\Rightarrow n = 4$$

Q80: The sum of possible values of x for

$$\tan^{-1}(x+1) + \cot^{-1}\left(\frac{1}{x-1}\right) = \tan^{-1}\left(\frac{8}{31}\right) \text{ is :}$$

(A) $-\frac{32}{4}$

(B) $-\frac{31}{4}$

(C) $-\frac{30}{4}$

(D) $-\frac{33}{4}$

Q80: The sum of possible values of x for

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(A) $-\frac{32}{4}$

(B) $-\frac{31}{4}$

(C) $-\frac{30}{4}$

(D) $-\frac{33}{4}$

Solution:

Taking tan both sides

$$\frac{(1+x)+(x-1)}{1-(1+x)(x-1)} = \frac{8}{31}$$

$$\Rightarrow \frac{2x}{2-x^2} = \frac{8}{31}$$

$$\Rightarrow 4x^2 + 31x - 8 = 0$$

$$\Rightarrow x = -8, \frac{1}{4}$$

$$\text{But at } x = \frac{1}{4}$$

$$LHS > \frac{\pi}{2} \text{ and } RHS < \frac{\pi}{2}$$

So only solution is $x = -8$

Q81: If the equation of the plane passing through the line of intersection of the planes $2x - 7y + 4z - 3 = 0$, $3x - 5y + 4z + 11 = 0$ and the point $(-2, 1, 3)$ is $ax + by + cz - 7 = 0$, then the value of $2a + b + c - 7$ is

Q81: If the equation of the plane passing through the line of intersection of the planes $2x - 7y + 4z - 3 = 0$, $3x - 5y + 4z + 11 = 0$ and the point $(-2, 1, 3)$ is $ax + by + cz - 7 = 0$, then the value of $2a + b + c - 7$ is

4.00

Solution:

Let equation of plane be $2x - 7y + 4z - 3 + \lambda(3x - 5y + 4z + 11) = 0$

This passes through $(-2, 1, 3)$

So, $-4 - 7 + 12 - 3 + \lambda(-6 - 5 + 12 + 11) = 0$

$$\Rightarrow \lambda = \frac{1}{6}$$

$\therefore$ Plane is $15x - 47y + 28z - 7 = 0$

$$\therefore a = 15$$

$$b = -47$$

$$c = 28$$

Q82: If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ is equal to

Q82: If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ is equal to

Solution:

$$|A| = -2 \Rightarrow |A|^4 = 16$$

$$A^{10} = \begin{bmatrix} 2^{10} & 2^{10} - 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1024 & 1023 \\ 0 & 1 \end{bmatrix}$$

$$2A = \begin{bmatrix} 4 & -6 \\ 0 & 4 \end{bmatrix}$$

$$\text{adj}(2A) = \begin{bmatrix} -2 & -6 \\ 0 & 4 \end{bmatrix}$$

$$\text{adj}(2A) = -2 \begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}$$

$$(\text{adj}(2A))^{10} = 2^{10} \begin{bmatrix} 1 & 3 \\ 0 & -2 \end{bmatrix}^{10}$$

$$= 2^{10} \begin{bmatrix} 1 & -(2^{10} - 1) \\ 0 & 2^{10} \end{bmatrix}$$

$$= 2^{10} \begin{bmatrix} 1 & -1023 \\ 0 & 1024 \end{bmatrix}$$

$$A^{10} - (\text{adj}(2A))^{10} = \begin{bmatrix} 0 & 1023(1 + 2^{10}) \\ 0 & 1 - (1024)^2 \end{bmatrix}$$

$$|A^{10} \text{adj}(2A)^{10}| = 0$$

Q83: The minimum distance between any two point P_1 and P_2 while considering point P_1 on one circle and point P_2 on the other circle for the given circles' equations

$$x^2 + y^2 - 10x - 10y + 41 = 0$$

$$x^2 + y^2 - 24x - 10y + 160 = 0 \text{ is } \dots\dots\dots .$$

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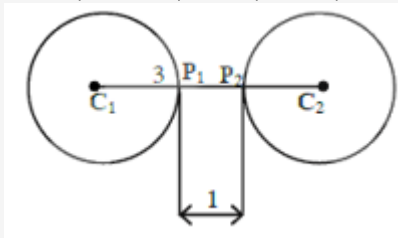
$$x^2 + y^2 - 24x - 10y + 160 = 0 \text{ is } \dots\dots\dots$$

1.00

Solution:

$$S_1 : (x - 5)^2 + (y - 5)^2 = 9, \text{ centre } (5, 5), r_1 = 3$$

$$S_2 : (x - 12)^2 + (y - 5)^2 = 9, \text{ centre } (12, 5), r_2 = 3$$



$$(P_1 P_2)_{\min} = 1$$

Q84: If $(2021)^{3762}$ is divided by 17, then the remainder is

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4.00

Solution:

$$\begin{aligned}(2021)^{3762} &= (2023 - 2)^{3762} = \text{multiple of } 17 + 2^{3762} \\&= 17\lambda + 2^2 (2^4)^{940} \\&= 17\lambda + 4(17 - 1)^{940} \\&= 17\lambda + 4(17\mu + 1) \\&= 17k + 4; (k \in I) \\&\text{Remainder} = 4\end{aligned}$$

Q85: Let there be three independent events E_1 , E_2 and E_3 . The probability that only E_1 occurs is α , only E_2 occurs is β and only E_3 occurs is γ . Let 'p' denote the probability of none of events occurs that satisfies the equation $(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$. All the given probabilities are assumed to lie in the interval (0, 1).

Then, $\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3}$ is equal to

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Then, $\frac{\text{Probability of occurrence of } E_1}{\text{Probability of occurrence of } E_3}$ is equal to

6.00

Solution:

Let x, y, z be probability of E_1, E_2, E_3 respectively

$$\Rightarrow x(1-y)(1-z) = \alpha \Rightarrow y(1-x)(1-z) = \beta \Rightarrow z(1-x)(1-y) = \gamma \Rightarrow (1-x)(1-y)(1-z) = P$$

Putting in the given relation we get $x = 2y$ and $y = 3z$

$$\Rightarrow x = 6z$$

$$\Rightarrow \frac{x}{z} = 6$$

Q86: The maximum value of z in the following equation, $z = 6xy + y^2$ where $3x + 4y \leq 100$ and $4x + 3y \leq 75$ for $x \geq 0$ and $y \geq 0$ is

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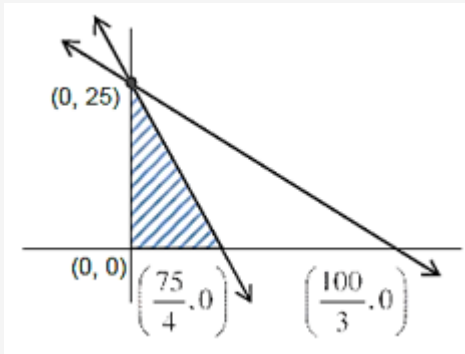
904.00

Solution:

$$x \geq 0, y \geq 0$$

$$4x + 3y \leq 75$$

$$3x + 4y \leq 100$$



The point for maximum value of z cannot lie inside the shaded triangle as for any point $A(x_0, y_0)$ inside the shaded region, we can have a point higher than A (or to the right of A), where z is larger than at A .

So the required point must lie on the boundary point, i.e. on the line $4x + 3y = 75$

$$\text{Using } x = \frac{(75 - 3y)}{4}$$

$$z = \frac{6y(75 - 3y)}{4} + y^2$$

Now maximum y is obtained at $-\frac{b}{2a}$ of this quadratic expression in y

$$\text{So } y = \frac{900}{56}, \text{ and } x = \frac{375}{56}$$

$$z_{\max} = 904$$

Q87: If $f(x) = \sin \left(\cos^{-1} \left(\frac{1-2^{2x}}{1+2^{2x}} \right) \right)$ and its first derivative with respect to x is $-\frac{b}{a} \log_e 2$ when $x = 1$, where a and b are integers, then the minimum value of $|a^2 - b^2|$ is

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Q87: If $f(x) = \sin\left(\cos^{-1}\left(\frac{1-2^{2x}}{1+2^{2x}}\right)\right)$ and its first derivative with respect to x is $-\frac{b}{a}\log_e 2$ when $x = 1$, where a and b are integers, then the minimum value of $|a^2 - b^2|$ is

.....

481.00

Solution:

$$\begin{aligned}
 2^x &= \tan \theta \quad \theta \in \left(0, \frac{\pi}{2}\right) \\
 &= \sin\left(\cos^{-1}\left(\frac{1-\tan^2 \theta}{1+\tan^2 \theta}\right)\right) \\
 &= \sin\left(\cos^{-1}(\cos 2\theta)\right) \\
 &= \sin(2\theta) \\
 &= \sin(2\tan^{-1}(2^x)) \\
 \therefore \frac{df}{dx} &= \cos(2\tan^{-1}(2^x)) \cdot 2 \cdot \frac{1}{1+2^{2x}} \cdot 2^x \log 2 \\
 \left.\frac{df}{dx}\right|_{x=1} &= \cos(2\tan^{-1}2) \cdot \frac{4}{5} \log 2 \\
 &= \frac{1-\tan^2(\tan^{-1}2)}{1+\tan^2(\tan^{-1}2)} \cdot \frac{4}{5} \log 2 \\
 &= \frac{-12}{25} \log 2 \\
 \text{So, } |a^2 - b^2| &= |625 - 144| = 481
 \end{aligned}$$

Q88: If $[\cdot]$ represents the greatest integer function, then the value of

$$\left| \int_0^{\sqrt{\frac{\pi}{2}}} [x^2] - \cos x \, dx \right| \text{ is } \dots\dots\dots$$

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$$\left| \int_0^{\sqrt{\frac{\pi}{2}}} [x^2] - \cos x \, dx \right| \text{ is } \dots\dots\dots$$

1.00

Solution:

$$\begin{aligned} & \int_0^{\sqrt{\frac{\pi}{2}}} ([x^2] + [-\cos x]) \, dx \\ &= \int_0^{\sqrt{\frac{\pi}{2}}} [x^2] \, dx + \int_0^{\sqrt{\frac{\pi}{2}}} (-1) \, dx \\ &= \int_0^1 0 \, dx + \int_1^{\sqrt{\frac{\pi}{2}}} 1 \, dx - \int_0^{\sqrt{\frac{\pi}{2}}} 1 \, dx \\ &= \sqrt{\frac{\pi}{2}} - 1 - \left(\sqrt{\frac{\pi}{2}} - 0 \right) \\ &= -1 \end{aligned}$$

Q89: If the function $f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$ is continuous at each point in its domain and $f(0) = \frac{1}{k}$, then k is

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6.00

Solution:

$$\begin{aligned} \frac{1}{k} &= \lim_{x \rightarrow 0} \frac{2 \sin\left(\frac{\sin x + x}{2}\right) \sin\left(\frac{x - \sin x}{2}\right)}{x^4} \\ &= \lim_{x \rightarrow 0} \frac{(\sin x + x)(x - \sin x)}{2x^4} = \left(\lim_{x \rightarrow 0} \frac{\sin x + x}{2x} \right) \left(\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} \right) \\ &= 1 \times \lim_{x \rightarrow 0} \frac{1 - \cos x}{3x^2} = \frac{1}{6} \\ k &= 6 \end{aligned}$$

Q90: If $\vec{a} = \alpha\hat{i} + \beta\hat{j} + 3\hat{k}$,

$\vec{b} = -\beta\hat{i} - \alpha\hat{j} - \hat{k}$ and $\vec{c} = \hat{i} - 2\hat{j} - \hat{k}$

Such that $\vec{a} \cdot \vec{b} = 1$ and $\vec{b} \cdot \vec{c} = -3$, then $\frac{1}{3} \left(\left(\vec{a} \times \vec{b} \right) \cdot \vec{c} \right)$ is equal to

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.....

Solution:

$$\vec{a} \cdot \vec{b} = 1$$

$$\Rightarrow -\alpha\beta - \alpha\beta - 3 = 1$$

$$\Rightarrow 2\alpha\beta = -4$$

$$\Rightarrow \alpha\beta = -2$$

$$\vec{b} \cdot \vec{c} = -3$$

$$\Rightarrow -\beta + 2\alpha + 1 = -3$$

$$\Rightarrow 2\alpha - \beta = -4$$

$$(2\alpha + \beta)^2 = (2\alpha - \beta)^2 + 8\alpha\beta$$

$$(2\alpha + \beta)^2 = 16 - 16 = 0$$

$$\Rightarrow 2\alpha + \beta = 0$$

$$\Rightarrow \alpha = -1, \beta = 2$$

$$\frac{1}{3} \left((\vec{a} \times \vec{b}) \cdot \vec{c} \right) = \frac{1}{3} \begin{vmatrix} -1 & 2 & 3 \\ -2 & 1 & -1 \\ 1 & -2 & -1 \end{vmatrix}$$

$$R_1 \rightarrow R_1 + R_3$$

$$= \frac{1}{3} \begin{vmatrix} 0 & 0 & 2 \\ -2 & 1 & -1 \\ 1 & -2 & -1 \end{vmatrix}$$

$$= \frac{1}{3} \times 2 \times 3 = 2$$